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Design and Evaluation of a Versatile Text Input Device for Virtual and Immersive Workspaces

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ABSTRACT

While headsets dedicated to mixed reality are increasingly considered to virtualize workspaces in the office, the keyboard remains mostly unchanged and nevertheless is still envisioned as an interface for common professional activities heavily demanding in text entry. However, unlike headsets, a keyboard is intrinsically limited to immobile context during its use, due to the need for physical support. In this article, we investigate a radical change in the structural shape of the keyboard, switching from the plane to a cube, allowing users to fully benefit from their whole environment by entering text in a much wider variety of context. We designed a cubic layout based on the QWERTY and results from two user studies to define the best holding position as well as fingers preferences and reachability to the cubic shape. Then, we conducted two subsequent user studies and one longitudinal case study to compare and evaluate typing skill transfer from the keyboard, learning curve, fingers usage, usability, and workload in various contexts. The cubic-shaped text entry device finally showed its versatility with users having similar performances while standing with a mixed-reality headset or sitting in front of a laptop, and an input speed from an expert reaching 55.1 wpm.

KEYWORDS

augmented reality; headset; text entry; cube; layout; design; evaluation

1. Introduction

With the increasing availability of augmented and mixed reality headsets, their contexts of use are becoming more diverse, but the text entry activity remains ubiquitous no matter the cases, whether it be, to enter short credentials, keywords, sending messages or annotate and even compose large documents, forms, spreadsheets and presentations. This is particularly true for most of the tasks done in an office workspace, where the workers are usually indoor and regularly engage in extensive text entries. In this context, headsets are increasingly considered to virtualize the office workspace, totally or partially immersive, for instance, to offer adaptive and wider displays while getting rid of the physical ones (Mcgill, Kehoe, Freeman, & Brewster, 2020), enabling, or at least easing, work everywhere (Grubert, Ofek, et al., 2018). However, the benefit of the headsets does not stop here, all the extended realities offered by these devices

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are inherently spatial, the virtual content is placed anywhere and sensed in a three-dimensional way, regularly inviting the users to physically move around the place to simultaneously manipulate virtual entities and enter text, thus including but also going beyond the seated virtual workspace. In this regard and for such an environment, several scenarios are already concerned and many more envisioned to take advantage of the headsets, most of them requiring a relatively large amount of text entry, for example, designing virtual full-scale models, navigating inside visually large three-dimensional databases, attending interactive real meeting rooms blended with virtual avatars, preparing augmented and virtual presentations, or simultaneously developing, coding and testing mixed reality applications.

Text entry is a very active field of research and a plethora of interfaces dedicated to the headsets have been explored, such as a virtual mid-air keyboards based on the movements of a specific body part, e.g., hands (Dudley, Vertanen, & Kristensson, 2018), head (Yu et al., 2017) or eyes (Rajanna & Hansen, 2018), as well as interfaces based on external devices such as dedicated controllers (Speicher, Feit, Ziegler, & Krüger, 2018), smartphones (Y. R. Kim & Kim, 2016), smartwatches (Ahn, Heo, & Lee, 2017), gloves (Whitmire et al., 2017) or rings (Gupta, Ji, Yeo, Quigley, & Vogel, 2019). But none of them reached the performance of the traditional physical keyboard, and since this device is currently by far the most used text entry interface in an office workspace, naturally, it has also been explored with headsets in several studies (Grubert, Witzani, et al., 2018; Knierim, Schwind, Feit, Nieuwenhuizen, & Henze, 2018; Walker, Li, Vertanen, & Kuhl, 2017). However, if intensive text entry relies on the traditional physical keyboard, in its current form, the users will remain relatively immobile while using it, often seated and bounded to a desk, mostly due to the need of a flat support, and thus, annihilating or highly limiting the spatial benefit of using augmented and mixed reality headsets. While the virtual keyboard on a smartphone offers mobility, it remains used mostly to light-text entries (fewer than 100 characters) for short and instant messaging (Li & Myers, 2021). For extensive (heavy) text entries in professional contexts, this interface still lacks performance and flexibility required by many workers, e.g., expertise in the use of multiple fingers, direct access to special characters, eye-free capability from tactile feedback, shortcut commands by combination (Le et al., 2020). Besides, an increase of two-thumb text entries (from smartphone or mini-keyboard) could worsen an already-known health issue, *texting thumb pain* (J. Chang, Choi, Tjolleng, & Jung, 2017; Williams & Kennedy, 2011).

To overcome these constraints, this article presents a novel interface by changing the structural form of the keyboard, switching from the traditional two-dimensional plane (or *board*) to a three-dimensional shape inspired by the vertical keyboard ergonomics, a cube, therefore, giving its near-generic name, *keycube*. A *keycube* aims to offer performance, flexibility, and familiarity similar to that of a traditional physical keyboard while bringing the advantages of mobility to the workplace and greater freedom of use in terms of body posture.

Our implementation of a *keycube* has 80 keys equally disposed by 4x4 matrices on 5 faces, with hardware specifications mostly based on international standard and ergonomics studies from the literature. Nonetheless, inspired by the user-driven development, two user studies were first conducted to slightly refine the prototype and design a cubic layout.

Study 1 (p=31) investigated how a cubic shape would be held while being used to enter text. The results revealed that one position is largely preferred, called *diagonal*; only two edges of the cube are in contact with the palms of the hands when they cover the device. Qualitative feedback was also collected.

Study 2 (p=22) determined which areas of the cubic shape are considered reachable and preferred by the users while holding a cube with the diagonal position.

Based on the results from the first two studies, the literature, and the design philosophy, the keycube adopted a cubic layout inspired by the QWERTY. Then, two subsequent user studies have been conducted to assess the text entry performance of the keycube in different contexts.

Study 3 (p=24) comparatively determined the progressive performance of different cubic layouts with a keycube, in a seated context, based on the following variables upon initial exposure to the device: input speed, error rate, workload, and usability. The outcomes revealed that the QWERTY-based cubic layout offered an overall advantage due to knowledge transfer, resulting in a steeper learning curve with novice users reaching up to 19 words per minute (wpm) within 100 minutes of practice during the same half-day, representing 30% of their input speed with traditional keyboard.

Study 4 (p=9) measured the performance of a keycube connected to an augmented reality headset while the users were standing, without the help of any physical support. The results showed that a keycube offered the same performance either seated or standing, and thus, confirming the versatile characteristic of this device. The second part of this study analyzed the fingers usage while typing with a cubic-shaped device (a keycube) comparatively to a plane-shaped device (a keyboard). The results revealed that the plane and cubic shapes already offered a similar load of fingers usage, despite a shift of used fingers (respectively from index and middle fingers, to index and thumbs).

Lastly, a longitudinal case study (p=1) was conducted and showed a highly progressive and promising possibility of a more experienced text entry with a keycube through a month of short sessions. The user reached 55.1 words per minute in less than four hours of accumulated practice.

All studies measuring text entry performances also include and present different baseline interfaces, such as traditional physical keyboards, virtual keyboard from the smartphones or head-then-hand gesture keyboards, not only for comparative purposes but also, more importantly, to validate our implementations and methodologies with known results from literature.

1.1. Outline

After reviewing the related work, this article presents the design criteria and philosophy of the keycube concept as well as the hardware details of a first prototype implementation. Then, it goes through the four user studies mentioned above, separated by the design presentation of the cubic layout. Finally, the case study is presented, and the article is concluded with a discussion of the limits but also opportunities and implications to design and deploy a keycube in workplaces as an input device for augmented and mixed reality headsets.

2. Related Work

In this section, previous works limited on text entry interfaces dedicated to headsets are first reviewed, then more specifically on typing devices, and lastly, on cubic-shaped interactive devices.

2.1. Text Entry Interfaces for Headsets

Since entering text with headsets is a very active research domain, previous works are reported by categories, except for specific additional details from major contributions. Besides, the text entry interfaces are reported independently of the targeted headsets, many solutions were deliberately addressing all of them and most of the latest virtual reality headsets already integrate sensors to also allow truly mixed reality experience; thus, they are sharing more similarities than differences from our perspective on this research.

A large amount of text entry interfaces are based on a virtual keyboard, usually QWERTY-based, floating up in the air in front of the user’s field of view, yet the interaction can be quite different. Some interfaces (M.-W. Chang, Chiueh, & Chang, 2014; Dudley et al., 2018; Ishii et al., 2017; Kwon et al., 2016; Speicher et al., 2018) will mostly use mid-air gestures of the arms, hands, and fingers to select and validate characters (or words), with the advantage to rely on dexterous members of the human body, without the need of external devices. Dudley et al. (2018) have shown that users, with a single hand, can achieve up to 17.75 wpm within a short period of time (less than two hours) by adding word prediction and auto correct. The main constraint of this kind of interaction is known as the *gorilla-arm syndrome (or effect)*, quickly causing arm fatigue and discomfort to the users while performing mid-air gestures. This issue has been studied, even compared with traditional keyboard usage (Hansberger et al., 2017), and countered only by using physical supports (chair armrest for the elbows), limiting at the same time the mobility advantage of this kind of interaction.

When both hands are unavailable (e.g., occupied with tools), which is less of a concern for virtual workspaces in offices or at home, the characters on the virtual floating keyboard can be selected and validated by moving a cursor with head (Speicher et al., 2018; Yu et al., 2017) or eyes (Rajanna & Hansen, 2018) movements. Both approaches must handle the *Midas touch problem* (Jacob, 1991), meaning that everything focused might be activated (such as virtual key, in the context of text entry). The most common method to alleviate this problem is to use dwell time as a threshold for validation (e.g., 400 milliseconds), which highly limits text entry performance, in particular the input speed. Another common method is multimodal, keeping the eyes or head movement to the characters selection while using another modality for the validation, such as pressing a button on an external device. In this configuration, Yu et al. (2017) showed that using head movements with an external device (in their case, a gamepad) users could reach up to 19.04 wpm by adopting word-level input from the gesture typing style. However, having to use the hands might limit the original benefit of interfaces based on head or eye movements.

External devices to the headsets have also been widely explored as a primary input method for text entry. For instance, still with the virtual floating keyboard, 6-DoF (*degree of freedom*) controllers dedicated to headsets have been used to select (by pointing) and validate the characters, either by pressing a button or by movement (Boletsis & Kongsvik, 2019; Chen, Wang, Guerra, Mittal, & Prakkamakul, 2019; Huang, Sun, Qin, Gao, & Qin, 2022; Y. Lee & Kim, 2017; Speicher et al., 2018). Using this technique combined with a word-gesture feature, Chen et al. (2019) reported on an expert’s input speed of 34.2 wpm.

Despite slight variations, most text entry interfaces based on gloves use pinch as input interactions, e.g., Bowman, Rhoton, and Pinho (2002); Fashimpaur, Kin, and Longest (2020); Jiang, Weng, Dongye, and Liu (2022); Kuester, Chen, Phair, and Mehring (2005); L. H. Lee et al. (2019); Whitmire et al. (2017)), usually the thumbs

press another part of the hand to both select and validate a character or word. These interfaces have the advantage to offer discreet interactions, with arms down, but the input speed remains relatively low. Whitmire et al. (2017) mapped the QWERTY layout along the fingers of both hands and reported on 16.5 wpm after ten 20-minute sessions.

Less common categories of external devices (with more variability among their implementations) have been explored as interfaces for headsets, usually for a relatively short text entry, for instance, wristband or smartwatches (Ahn et al., 2017; Prätorius, Valkov, Burghbacher, & Hinrichs, 2014), clothes (Belkacem, Pecci, Martin, & Faiola, 2019), rings (Fukumoto & Suenaga, 1994; Gupta et al., 2019; J. Kim, Delamare, & Irani, 2018), synthetic skin (Xu et al., 2020), pens (Bowman et al., 2002; González, Molina, García, Martínez, & González, 2009; Poupyrev, Tomokazu, & Weghorst, 1998). Most of these solutions tried to leverage on the QWERTY layout, even for those with limited physical surfaces, usually by following a two-step selection method on grouped letters. Few other solutions preferred to implement a chording technique, in this case, entering text is done by combining actions (such as fingers tap (Fukumoto & Suenaga, 1994) or touch (Belkacem et al., 2019)).

Speech, usually accompanied to another modality, has also been explored with headsets as a text entry interface (Adhikary & Vertanen, 2021; Bowman et al., 2002; Ghosh et al., 2020). This technique is incompatible in the context of intensive text entry and workspaces, particularly in shared offices, mostly to avoid overlapping with the already widely used and noisy human-to-human speech communication, in addition to the known issues such as potentially disclosing personal and confidential information or the cognitive cost of speech (Shneiderman, 2000).

Finally, and naturally, among the techniques taking advantage of an external device, the two most common text entry interfaces have been widely explored: physical keyboards (Bovet et al., 2018; Grubert, Witzani, et al., 2018; Hoppe, Otto, van de Camp, Stiefelhagen, & Unmüßig, 2018; Jiang et al., 2018; S. Kim, Kim, Kim, & Kim, 2004; Knierim et al., 2018; Lin et al., 2017; McGill, Boland, Murray-Smith, & Brewster, 2015; Otte et al., 2019; Pham & Stuerzlinger, 2019; Walker et al., 2017) and virtual keyboard on smartphones (Bai, Zhang, Yang, & Billingham, 2021; Boustila, Guégan, Takashima, & Kitamura, 2019; Chen et al., 2019; Y. R. Kim & Kim, 2016; Lu, Yu, Yi, Shi, & Zhao, 2017). Although mobility is not an issue with the smartphones, the average input speed is about 70% (at around 36.2 wpm (Palin, Feit, Kim, Kristensson, & Oulasvirta, 2019)) of the traditional keyboard (at around 52 wpm (Dhakal, Feit, Kristensson, & Oulasvirta, 2018)) and the current lack of haptic feedback from the touchscreen has been reported in the literature as a major constraint for users to adopt smartphones with intensive text entry activities (Le et al., 2020). Moreover, the smartphone typing performances are highly dependent on the user’s gaze toward the touchscreen, forcing the users to adopt specific postures, usually by looking down at the device, eventually causing neck musculoskeletal disorder and pain (Eitviki, Viriyarajanakul, & Redhead, 2018). For the previous attempts (Lu et al., 2017; Yang, Yu, Yi, & Shi, 2019) at giving eyes-free capability to the use of keyboard on smartphone, (1) the performances decreased (to 22.3 wpm), (2) the visual interface was transferred to another screen which still required users’ eyesight and visual attention to input text (e.g., to select words among a list or even to visual the whole keyboard), thus, falling short at the definition of eyes-free, and (3) the interfaces lacked any solution (neither were apparent) to enter out-of-dictionary words, such as commonly encountered: numbers, proper names, punctuations and other special characters. Finally, the smartphone keyboard is mostly limited to only one or two fingers (Palin et al., 2019),

which in addition to clearly lacking the potential of some human resources (in this case, fingers), is prone to muscular stress, *texting thumb pain* (J. Chang et al., 2017; Williams & Kennedy, 2011). Engaging even more with the smartphone to intensive text entry could only worsen these issues.

On the other hand, the traditional keyboard, which benefit from a much higher flexibility and performance than the smartphone keyboard (summarized in this comic¹), remains the most used text entry interface in the context of office workplace and many previous works focused on using it with virtual reality headsets in which the hands and the devices were hardly visible. Progress was made to the point that there is already little difference in performances by typing with or without wearing a headset (Pham & Stuerzlinger, 2019), which is expected for augmented or mixed reality headsets. Pham and Stuerzlinger (2019) also expressed our concern about the relative immobility of the users due to the constraining requirement for physical supports while using a traditional keyboard. They suggested wearing a tray attached in front of the user, large enough to support the device, thus, allowing people to stand with and move around.

2.2. *Typing Devices*

This part reports on typing devices (or techniques) with physical keys, sharing characteristics with the concept of keycube, going beyond the use of headsets.

First, the chording technique, presented in the previous part, was also implemented with dedicated physical devices, known as chord keyboards (Gopher & Raij, 1988; Noyes, 1983a; Rochester, Bequaert, & Sharp, 1978), allowing to type with a very limited number of keys while achieving high input speed, studies report on users reaching 35 wpm after 20 hours of practice. The Twiddler (Lyons, Plaisted, & Starner, 2004; Lyons, Starner, et al., 2004) is a highly mobile one-handed chord keyboard with comparable results. However, this interface suffers from substantially low novice performance and a steep learning curve, preventing users from adopting it (Bowman et al., 2002; González et al., 2009).

To provide access to disable users, the Half-QWERTY (Matias, MacKenzie, & Buxton, 1993), a one-handed typique technique was designed to facilitate skill transfer from two-handed typing. It was shown that QWERTY typists could achieve more than 40 wpm in less than 10 hours. For similar motivations, the Stick Keyboard (Green, Kruger, Faldu, & St. Amant, 2004) reduces the physical space requirement and capitalizes on user knowledge of traditional keyboards by mapping four rows of characters to a single row. Users achieved up to 22.5 wpm with a lexicon-based disambiguation technique, but 10.4 wpm by multi-tapping. Other works have explored typing techniques based on a highly reduced number of keys, from 1 to 5 (Baljko & Tam, 2006; I. S. MacKenzie, 2009; S. MacKenzie, 2002) but came with a decreased input speed of around 5 to 10 wpm and often implied the need of software optimization and disambiguation.

Compactness of the traditional keyboard was also explored without changing the layout but by reducing overall size, including its keys, usually known as compact or mini-keyboards. While this keyboard transformation brings mobility, it also comes with a drastic change of usage, the devices are now mostly used with thumbs only. Despite studies reporting theoretically (I. S. MacKenzie & Soukoreff, 2002b) and empirically (Clarkson, Clawson, Lyons, & Starner, 2005; Clawson, Lyons, Starner, & Clarkson, 2005) that users may achieve an input speed of 60 wpm, these devices are in a way

¹When I'm Back at a Keyboard, by Randall Munroe, published July 29, 2019 at <https://xkcd.com/2182>

similar to the text entry interface with a smartphone, thus, prone to the *texting thumb pain* (Gustafsson, Coenen, Campbell, & Straker, 2018) and lack combination possibilities for shortcut commands requiring more than two fingers, limiting the expertise required by many workers.

The traditional keyboard has already been modified in various ways to offer different advantages. For instance, an increase of comfort was pursued with the split keyboards, in which two separated parts allowed more freedom to the spacing and orientation of the hands (Honan, Serina, Tal, & Rempel, 1995; Rempel, 2008; Smith et al., 1998). Split technique have been used with RearType (Scott et al., 2010) by disposing 30 keys (around 30% of a traditional keyboard) on the reverse side of a tablet. Exploiting potential knowledge transfer from the QWERTY layout, users achieved 15 wpm within one hour of training. Similar results regarding input speed were achieved with the Back Keyboard (H. Kim, Row, & Lee, 2012) where physical size-reduced keys were placed on the backside of a mobile phone.

Studies have also explored different inclinations of the split keyboard, up to the vertical, while reporting less stress on muscles sensitive to repetitive strain injuries without decreasing much of the input speed after a week of use (Muss & Hedge, 1999; Van Galen, Liesker, & de Haan, 2007). As traditional, vertical keyboards, remain constrained to relatively immobile usage by needing physical support. Sharing our immobility concern over the use of a traditional keyboard with headsets, a split keyboard was vertically mounted on the left and right side of a virtual reality headset (Hutama, Harashima, Ishikawa, & Manabe, 2021). They reported on a preliminary user study with 3 participants, all touch typers, achieving 34.7 wpm after three days of training. However, this solution, forces the users to adopt a unique and tiring position for extensive text entries, with the arms up and bent toward the head. Finally, when the two parts of a vertical keyboard are joined, they can form a shape similar to a triangular prism, or more precisely, half of a cube split by a diagonal if the transverse plane angle is 90 degrees (K.-J. Lee et al., 2009). While both shapes would allow mobility, we believe the cube is easier than the triangular prism to grasp and offers simpler three-dimensional mental representation deemed useful for immersive environments.

2.3. Cubic-Shaped Devices

The cubic shape has become popular since Sheridan, Short, Laerhoven, Villar, and Kortuem (2003) explored and showed its natural affordance. Thus, many previous works have investigated a large variety of cubic shapes to create interactive devices from learning tools (Goh, Kasun, Tan, & Shou, 2012; Merrill, Sun, & Kalanithi, 2012; Oh & Gross, 2015; Terrenghi, Kranz, Holleis, & Schmidt, 2006) to displays (Pla & Maes, 2012; Stavness, Lam, & Fels, 2010).

Closer to the concept of keycube, were a great number of cubic devices also used for input, in particular for three-dimensional manipulations (Chakraborty et al., 2014; de la Rivière, Kervégant, Orvain, & Dittlo, 2008; Fröhlich & Plate, 2000; Ichida, Itoh, Kitamura, & Kishino, 2004; Lertsithichai & Seegmiller, 2002; Murakami & Nakajima, 1994; Roudaut et al., 2014), intuitively facilitating operations (such as selection, translation, rotation or scaling) commonly encountered with computer-aided design (CAD) applications. Most of the interactions were done by matching the physical manipulation of the device with a virtual entity (sometimes experienced in augmented or virtual reality) by integrating different sorts of sensors, such as inertial measure units or expandable and deformable structures (Lertsithichai & Seegmiller, 2002; Murakami

& Nakajima, 1994). Few cubic devices included a limited number of buttons (Fröhlich & Plate, 2000; H. Lee & Woo, 2010; Roudaut et al., 2014) or provided different functionalities and properties depending on their face (Lefeuvre et al., 2016; Matviienko, Horwege, Frick, Ressel, & Boll, 2016). Only Brun, George, and Gouin-Vallerand (2020) introduced a cubic device for text entry but without any design implementation or studies for immersive environments.

Other related contexts have been explored, for instance, controlling TV media and music, e.g., channel, volume, playlist (Alonso & Keyson, 2005; Block, Schmidt, Villar, & Gellersen, 2004) by rotating or shuffling the cubic device; querying databases (Camarata, Do, Johnson, & Gross, 2002; Langner, Augsburg, & Dachsel, 2014; Zeller & Scherffig, 2009) by rotating or combining a set of cubes; navigating menus or stories (Juan, Canu, & Giménez, 2008; H. Lee & Woo, 2010; Zhou, Cheok, & Pan, 2004) by rotating or even manipulating a foldable cube.

2.4. Summary

In the first part, a wide range of previous text entry interfaces dedicated to headsets were presented by category, the most promising being these based on in-air hand gestures or controllers. Most of the solutions, despite being suitable for specific contexts with short text entries (e.g., industrial, with one hand occupied or while being on the go, outdoor), did not offer an input speed, and equally important, a level of usability and flexibility (e.g., eye-free capability or number of fingers involved) that is reachable with the traditional keyboards in the context of virtual offices.

However, with a traditional keyboard, the users are bound to a relatively immobile context, usually seated in front of a desk, limiting some benefits of using headsets. Thus, in the second part, typing devices, mostly in the form of modified traditional keyboards were presented, either to increase comfort (mostly by changing the orientation and inclination of the hands) or to bring compactness and address the mobility issue by reducing the overall size of the device with two different approaches: reducing the number or the size of the keys, but in both cases, it came with drastic change of usage, and performance.

Finally, in the third part, previous work presented cubic shaped devices for a variety of contexts, some highly relevant to the characteristics of augmented and mixed reality experiences, such as three-dimensional manipulations. The versatile perspective of the cube encouraged us to explore this platonic solid for a very pervasive activity: text entry.

3. Prototype - Cubic Device for Text Entry

Our first implementation of the keycube concept is similar to a keyboard but with a cubic shape easy to grasp, to hold, and small enough to be covered by the hands. Multiple key matrices are disposed over the faces of the cube (Figure 1). As a prototype, this peripheral-type device must be connected to a computing host (such as mixed reality headsets or laptops) and require dedicated applications; therefore, we present in this section two parts: hardware and software.

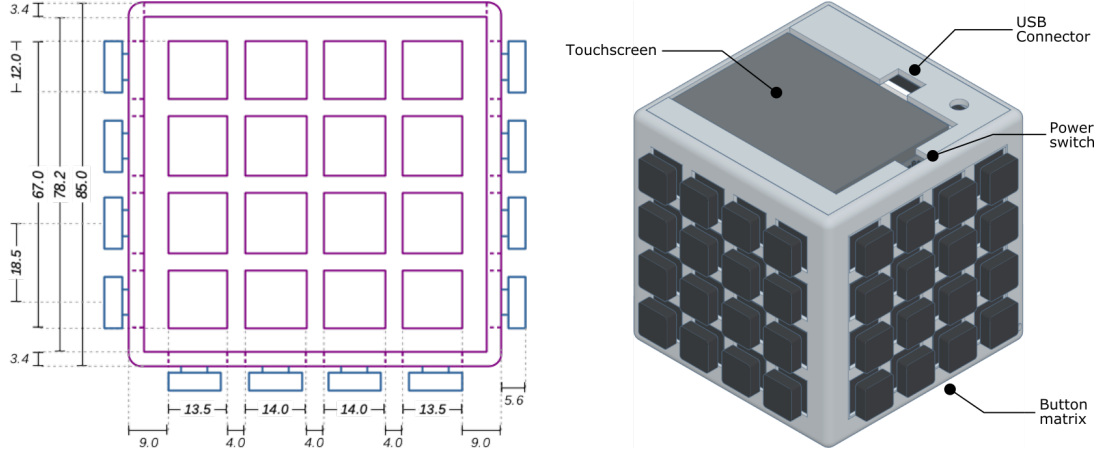


Figure 1. Structure size in millimeters (left) and 3D model (right) of a keycube prototype presented upside-down. Similar key matrices are placed on the hidden faces, an exploded view is provided in Appendix A.

3.1. Design Philosophy

The keycube is characterized by our main design considerations, which are (1) similarity to the traditional keyboard and (2) mobility (including great freedom of posture when using the device), in this regard we have established several criteria:

- Similar number of keys (with traditional keyboards)
- Similar key size
- Similar key spacing
- Freedom of use of the fingers (typing with one or more fingers)
- Freedom of gaze (typing while looking or not at the device)
- An overall size of the device considered small enough to be covered by one or both hands.

3.2. Hardware

Following these criteria and our inspiration from the vertical keyboard ergonomics and benefits (section 2.2), we have chosen to implement 16-key (4x4) ortholinear matrices on 5 sides of a cube, for a total of 80 keys, a quantity similar to the traditional physical keyboard on laptops, usually varying between 80 and 90 keys. The main structure of the device is 3D printed and measures 85x85x85 mm (Figure 1), but the extrusion of the keys increases the keycube total size up to 96.2x96.2x90.6 mm (LxWxH).

The number of keys and the size of the structure are both interlinked and were defined by starting with the recommended key size and spacing of traditional keyboards according to standards and a series of three studies (Madison et al., 2015; Pereira, Hsieh, Laroche, & Rempel, 2014; Pereira et al., 2013). The authors showed that a large key spacing 17-19 mm (calculated from their center) was significantly better for productivity and usability than a small spacing, 15.5-16 mm. Their results are consistent with the recommendations of the International Organization for Standardization (ISO) recommending a key spacing of 19+/-1 mm. These studies also showed that for the same spacing (of 17 mm), the precision (of text entry) was better with smaller key surfaces between 11 mm and 12 mm than with surfaces between 15 mm and 16 mm. We believe their results are beneficial for the keycube, since (1) the dimensions of the

hands and fingers remain the same and (2) the potential hand poses on the device are very similar to those already adopted with vertical keyboards. Therefore, the keys of the keycube have a 12 mm top surface and a spacing (both horizontally and vertically) of approximately 18.5 mm. Moreover, a 9 mm external space (border, including a rounding radius of 3.4 mm) has been added around each matrix to increase the area of the curved edges of the cube structure, reaching 16.5 mm (adult's fingertip width is around 16 mm to 20 mm) to facilitate grasping the device and to increase its comfort while being held in the palms of the hands (Sheridan et al., 2003). In such configuration, we discarded 9-key (3x3) matrices, due to the limited total number of keys (at 45), and 25-key (5x5) matrices due to the consequently overall device size considered too large (around 115x115x110 mm including the keys). We considered only square matrices (same number of rows and columns) over unspecific rectangular arrays (e.g., 3x4) to maintain symmetry, thus, allowing more freedom among users regarding the placement of the flat face while using the device. Since we did not find any user studies comparing ortholinear and staggered layouts, the keys of the keycube are arranged ortholinearly for the same reason (to maintain symmetry), besides practicality and aesthetics. Finally, before 3D printing the structure, multiple cubes were made from cardboard and tested internally to verify that the overall size of the prototype was small enough to be covered and easily manipulated with one or both hands.

The keys are metal dome switches with a travel distance around 0.5 mm. To limit unwanted and accidental key presses while handling the device, the actuation force of the keys is higher (between 1 N and 3 N, around 100-300 gf) than the recommendation (from the ISO) to the traditional keyboard (between 0.5 N and 0.8 N).

The key switches are mounted on custom prototype printed circuit boards (PCB). Eighty (80) diodes have been assigned to each of the keys, in order to eliminate common problems related to the matrices when the user performs simultaneous presses: *masking* (one or more pressed keys are not recognized) and *ghosting* (one or more keys are recognized without being pressed). The keycube has virtually a n -key rollover (NKRO), meaning that each key will be detected correctly no matter how many keys are pressed simultaneously.

The current prototypes have face-distributed color keycaps (red, green, blue, yellow, and white) to aid users' three-dimensional mental representations of the device. Transparent, movable, and optional key covers allow the characters to be inscribed on each of the keys while facilitating their switching for layout customization. The face without keys was implemented in order to:

- Give a distinctive tactile cue by breaking one axial symmetry, the users are then able to recognize for instance the top from the bottom, without having to look at the device.
- Have a convenient (flat) face to put down the device without pressing any keys.
- Include a tactile surface.

Beyond the keys, the keycube prototype includes other major components:

- A touch screen to simulate a pointing device or to have customizable virtual buttons (and other controls) that vary according to the host's current application, for instance, as shortcuts to enter emoticons or edit text (the style).
- An inertial unit (6 degrees of freedom with an accelerometer and gyroscope) to recognize the movements of the device.
- A vibrating motor to exploit additional tactile feedback during augmented, mixed, or virtual reality experiences (for instance, to simulate a collision be-

tween the keycube and a virtual entity or to give interaction feedback while using the touch screen).

- A Bluetooth module for a wireless connection (a Micro-B USB port remains available on the flat face of the device for a wired connection).

Finally, the device includes a (rechargeable) battery, a slide switch (power) and a microcontroller (ATmega2560 integrated on an Arduino board), for an overall weight of 430 grams. Due to the 3D printed nature of the structure, and its use of off-the-shelf components (shown in the exploded view, Appendix A), the device can be easily replicated or modified, for instance to add various components, such optical sensors or trackballs (to provide different point-and-click interaction modalities). In that regard, we also openly released the 3D models in the supplemental online material.

3.3. Software

The prototype is not yet *plug and play* (PnP) and does not follow any Human Interface Device (HID) specification; thus, the software part consists of both a program running on the keycube microcontroller and a specific application running on the host devices.

- *Microcontroller*. To compensate for the limited number of input/output (IO) pins allowed to read the electrical signals of the keys (the latter being more numerous), we have implemented a multiplexing system (executed at around 1.7 kHz). The program also manages to send information events by serial communication to the host devices in the form of an alphanumeric code, both by Bluetooth and universal serial bus (USB).
- *Host*. A companion application (compatible with Windows and macOS) has been developed with Processing (Figure 2, left) to (1) receive the alphanumeric codes of the keycube, (2) interpret them such as pressing or releasing keys and (3) simulate these events to the operating system. Besides, the application includes an optional *viewer*, which represents a symbolic and interactive visual feedback of a keycube displayed always-on top of other applications (example in Figure 2, right), highlighting key states (pressed or released) in real time. By adding exteroceptive cues (spatial-visual), the *viewer* may be an important feature regarding the user learning curve of a keycube (Fleishman & Rich, 1963). This application may be ported as Unity and Unreal Engine packages including a three-dimensional *viewer* to be integrated in most augmented, mixed, or virtual reality headsets.

Finally, the application allows a complete customization (mapping characters to any keys) of the cubic layout by the users, which includes the possibility to save them and load alternative predefined layouts.

With respect to the last point, particularly for designing a default predefined layout, we conducted two user studies: (1) preferred holding position, (2) finger-to-key preference and reachability.

4. User Study 1: Preferred Holding Position While in Use

The goal of this exploratory study was (1) to identify the most appropriate way to hold a cubic-shaped device while being used in the context of text entry, in order to design the cubic layout, and (2) to receive qualitative feedback from the keycube

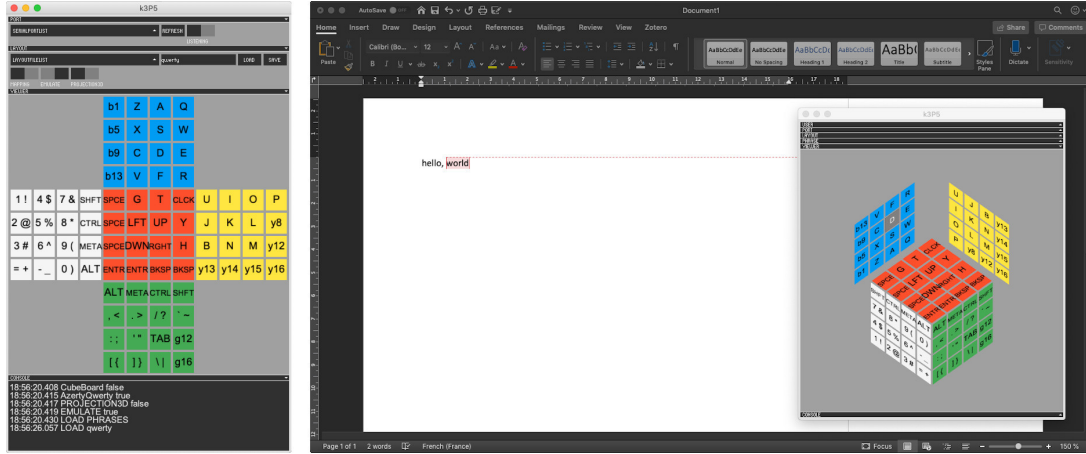


Figure 2. Companion application allowing customization, saving, and loading of cubic layouts (left) and screenshot of the keycube viewer above a traditional word processor application (right).

implementation. This first step was essential since depending on the hands and device positions chosen by the users, the fingers reachability and preferences to the keys will be different and therefore the placement of each character.

4.1. Participants and Materials

A total of 31 participants (2 left-handed; 15 women; 23-53 years old, mean 31) were recruited mostly from the university campuses (*Le Mans Université* in France and *Université TÉLUQ* in Canada). A neutral keycube was used, the transparent covers that support character inscriptions were removed from the device. Each key corresponded to an alphanumeric code (meaningless to the participants) and constructed as such: first letter of their color followed by a number from 1 to 16. The keycube was connected via Bluetooth to a laptop with a custom application which included an interactive two-dimensional cubic layout (similar to Figure 2, left) visually highlighting each key press and a text area transcribing their alphanumeric code.

4.2. Procedure

After being introduced to the study and answering a profiling survey, the participants freely manipulated a keycube during 5 minutes with the instructions: (1) to imagine using the prototype in the context of text entry for the whole experiment, (2) to try as many different holding positions as possible and (3) to follow a think-aloud protocol (while being video recorded). The participants were free to stand, move or even use a table as support. Then, to make sure all participants had explored enough positions, they were given a minimal set of different positions to try, related to either body posture, arms posture, or position of the device itself (Figure 3). Finally, participants had to choose their preferred positions to hold the device while simulating text entry, both in standing and sitting posture (pictures were taken). Finally, participants were given a questionnaire with open questions for potential improvements of the prototype. Each experiment lasted around 35 minutes.

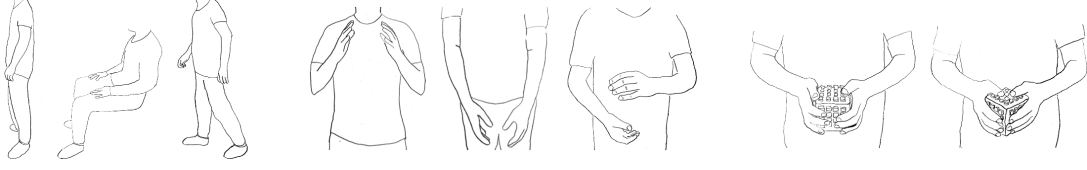


Figure 3. Minimal set of positions (from left to right) regarding the body (standing, sitting, walking), the arms (bent, released, upside-down) and the device (front, diagonal).

4.3. Results

The results came from questionnaires, video recordings of the manipulation, photographs of the preferred positions and the instructor’s notes taken from spontaneous feedback received during the experiments.

4.3.1. Preferred Holding Position While in Use

Two major holding positions to use the device emerged from the study: (1) *diagonal* and (2) *front* (Figure 4, respectively left and right), in both cases, the device was held with two hands to use more fingers, forearms slightly bent and the touch screen facing down. Two strategies were adopted to prevent the device from falling, either by placing the little fingers under the bottom face or by increasing the pressure of the hand palms against the lateral edges of the device (only possible with the *diagonal* position, otherwise many keys from the lateral faces would be accidentally pressed in the *front* position). We believe the presented interactive two-dimensional cubic layout (Figure 2, left) could have created a bias, making it easier to mentally map the real key to the layout when holding the device in front position (right in Figure 4). Therefore, during the free manipulation session, the *front* position was always discovered by the participants themselves (100%), but only 4 times (13%) for the *diagonal* position. However, after trying the minimal set of positions (Figure 3), this *diagonal* position was ultimately preferred by 77.4%, mostly because it offered more directly reachable keys by the fingers and a higher general comfort. Furthermore, six participants (19%) used a different holding position between the standing and sitting posture, 25 participants (81%) chose the diagonal position when sitting and 23 (74%) when standing. Three participants preferred to place the touch screen on top, more reachable by the thumbs (for example, to simultaneously perform point and click interactions) and judged in this case the keys of the bottom face reachable enough by the little fingers.

4.3.2. General Feedback

Five participants (16%) found the keycube interestingly easier to use while standing or walking than being seated. We believe this is due to the position adopted while moving, the arms relaxed, the bottom face of the keycube came against the body, and previous strategies to hold the apparatus are no longer necessary, naturally the whole fingers (or just the tips) are enough to prevent the device from falling (Figure 5, left).

Eleven participants (36%) used the keycube with one hand, either to type or by pretending it could act as a pointing device ($n=7$; 23%) by sliding it on the table or on their body to simulate a moving cursor. Due to its small form factor (85 mm edges size), all participants easily handled the keycube with one hand at some point, just passing it from one hand to another and to move around with it. The users can perform all direct interactions requiring one hand (e.g., pointing and clicking to move

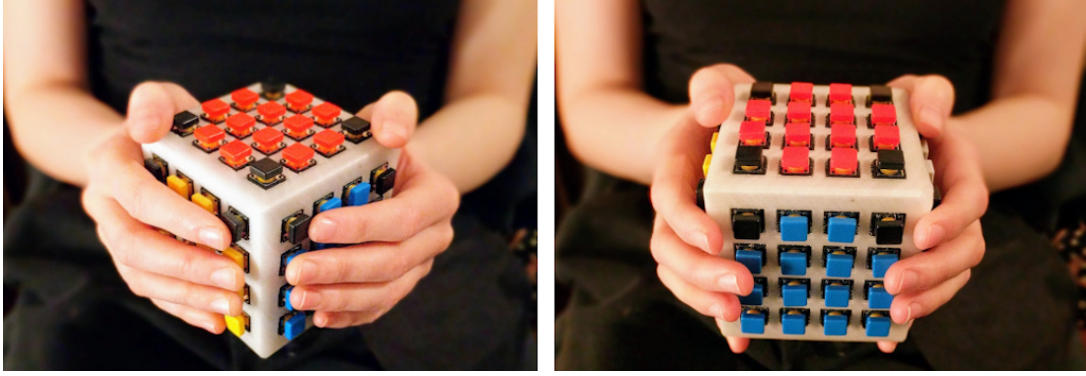


Figure 4. Preferred holding positions while using the device, *diagonal* on the left (chosen at 77.4%) and *front* on the right.

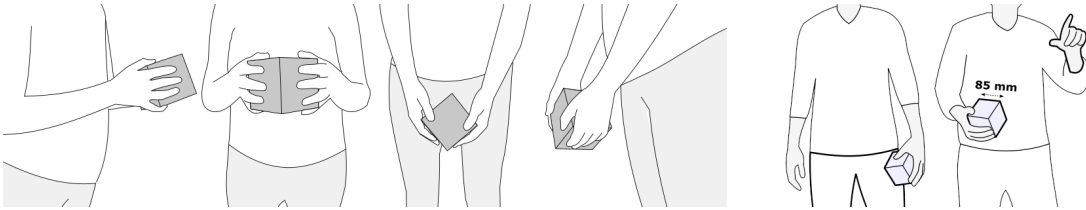


Figure 5. Holding position while using the device with the forearms bent and little fingers to prevent falling or with the arms relaxed, the device held by all the fingers (left). Holding the device with one hand to use it against the body or to perform other interactions (right).

the caret while editing text) and more complex but less common interactions requiring two hands could actually benefit from using the keycube’s other characteristics such as the inertial measurement unit for 3D manipulations. The colors were appreciated by six participants (19%), helping them to define dedicated zones. In that regard, two participants described the device as ideally playful for children and two others noted a de-stressing effect, after vigorously pressing many keys. Two participants thought about using a keycube as a way to play sports or relaxing games (by moving the device up in the air) and six (19%) compared the device to a musical instrument, in particular an accordion. As a result of its portability characteristic, the keycube has been seen as a device that encourages sharing and collaborating with coworkers and friends, or on the contrary, one participant appreciated the keycube proximity to the body and the “*personal intimate relation offered by the device.*”

Finally, participants also pointed out different potential upgrades to the device characteristics, such as slightly decreasing the weight (n=12; 39%) and the size (n=5; 16%), which asserted our choice to limit at 4x4 matrices. They reported the need to have an additional physical support (n=9; 29%) or a way to tie the device to the hands or body, e.g., a strap (n=4; 13%), mainly due to the fear of dropping the keycube (n=8; 26%) but also with the intention of easily freeing both hands while keeping the device. Five participants (16%) wanted wider support areas on the device itself (along the edges) or even fewer keys (n=6; 19%) in response to self-reported accidental key press during manipulation (n=11; 36%).

5. User Study 2: Finger-to-Key Preference and Reachability

To design the cubic layout, a second user study was conducted to identify and measure the preferred and reachable key areas from the fingers when the users hold a cubic-shaped device in the preferred position (*diagonal*) defined in the previous user study.

5.1. Participants and Materials

A total of 22 participants (2 left-handed; 14 women; 23-43 years old, mean 31) were recruited mostly from the university campuses (*Le Mans Université* in France and *Université TÉLUQ* in Canada), 13 participated in the previous study. The same material was used (a neutral keycube connected to a laptop via Bluetooth), but a new application was developed and showed an interactive cubic layout in an orthogonal axonometric view (similar to Figure 2, right) to ease the mental representation of the device when it is used in the *diagonal* position. A text area and symbolic hands accompanied the visual interface to guide the participants.

5.2. Procedure

After being individually introduced to the study and answering a profiling survey, participants were instructed to sit and hold the device in *diagonal* position for the whole experiment. The blue and yellow faces were arbitrarily chosen to be external while the white and green faces were internal (facing the user's body). The instructor made sure they respected this rule. The keys of the prototype implementation were used to identify preference and reachability areas of the cubic shape. Therefore, during the first part of the study, finger by finger, participants had to subjectively report which keys are *easily reachable* (usually straight at the tip of the finger or around) or *reachable with effort* (such as slight hand or finger movement) by pressing them and telling the alphanumeric code to the instructor. All the non-reported keys were considered *unreachable* by the participant. The hand and finger order among participants was counterbalanced to avoid some potential bias due to practice over time. For the second part of the experiment, the instructor said an alphanumeric code to the participants, which in return reported the preferred finger to press the corresponding key. All keys of the prototype (N=80) were randomly given to avoid a potential succession bias from keys placed next to each other, making sure the participants always held the device similarly without moving their hands (which would otherwise have been induced by a specific sequential keys order). A much sought-after consequence since the preferences must reflect an expertise, which is inversely correlated to the hands movement with a traditional keyboard, the less the hands move, the better are the performance (Feit, Weir, & Oulasvirta, 2016). Each experiment lasted an average of 48 minutes.

5.3. Results

Based on the collected data, this part present the reachability and preference areas of a cubic shape related to the users' fingers.

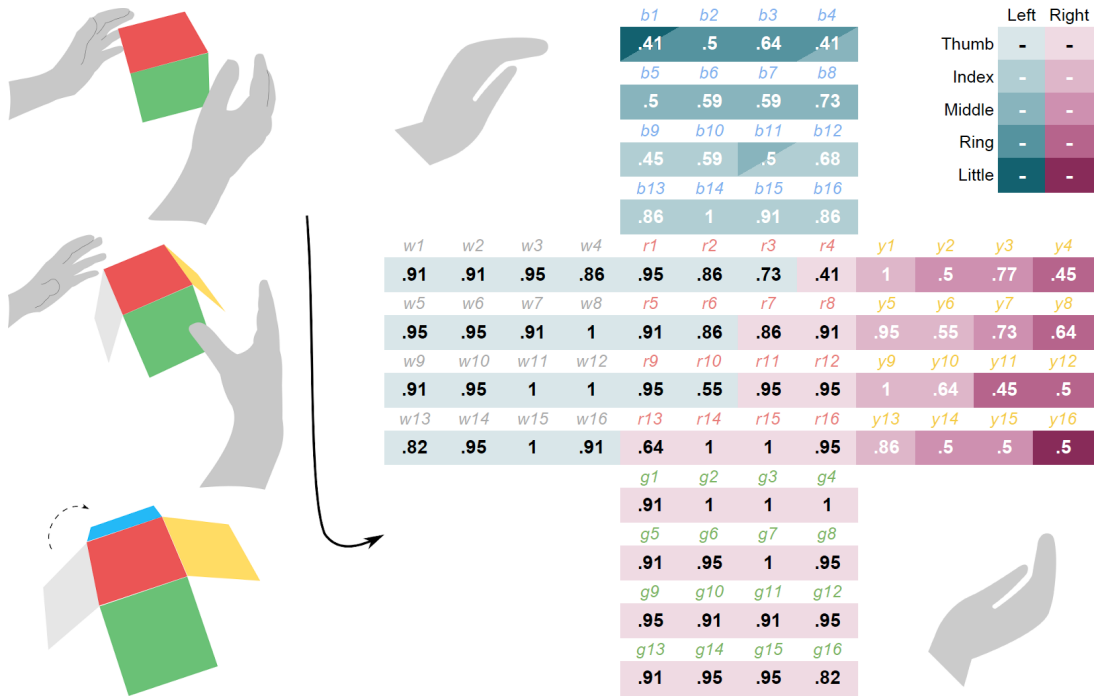


Figure 6. Finger preferences cartography including the confidence score (selection ratio) per key.

5.3.1. Finger-to-Key Preferences

Figure 6 shows a cartography of the keycube finger-to-key preferences. For each key, the most chosen finger among all participants was selected (reported by color code) and a confidence score calculated as its ratio varying theoretically between 0.14 (the preferred finger has been chosen by only 3 participants) and 1.0 (the same finger has been chosen by every participant). The alphanumeric codes of the keys are reported above the score in the color of their corresponding face. Three keys (b1, b4, b11) are bicolor because two fingers received the same preference score. The thumbs have the widest preference areas, dividing the red face (central in Figure 6) in half as well as occupying the white and green internal faces, while the little fingers are preferred only for a single touch each (b1 and y16).

5.3.2. Finger-to-Key Reachability

The answers to the reachability question have been pondered as such: 3 points for the keys considered as *easily reachable*, 1 point for *reachable with effort*, 0 points for *unreachable*. The total reachability score per key is presented in the left part of the Figure 7 and is the sum among all fingers and all users. Naturally, the most reachable areas are those close to the fingertips, needing less effort, in particular around the middle fingers, because the keys can also be easily pressed by the index and ring fingers.

Some keys have received a modest overall reachability score (around 80), but have a high reachability score only for a few fingers (for instance, the thumbs with the red face in the center, from r1 to r16), while other keys have a modest total reachability score without being strongly reachable with any of the fingers (for instance the key areas close to the edges of the cube in contact with the palms of the hands, b1, b5,

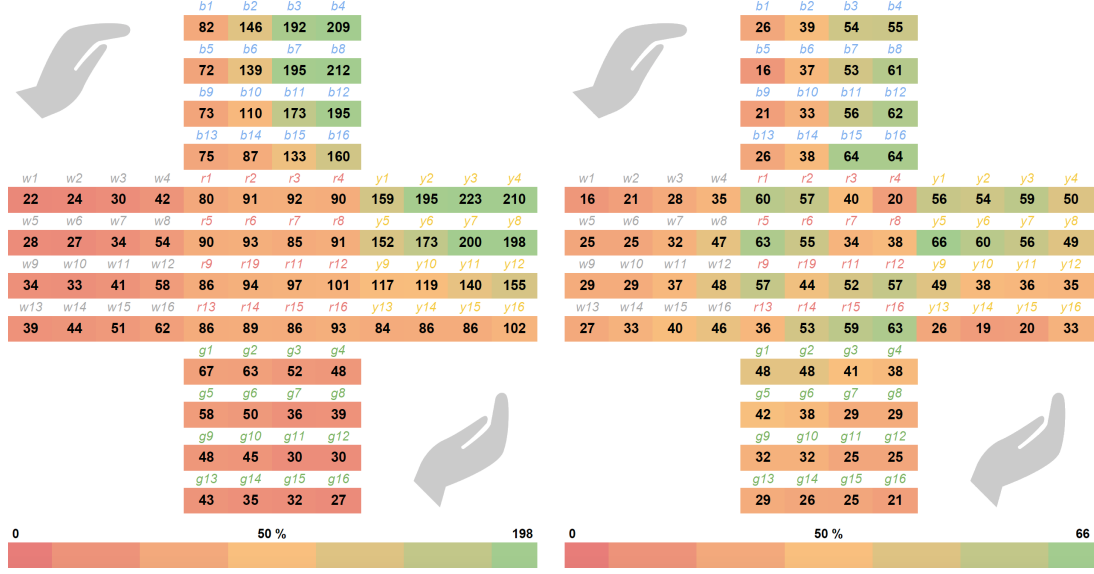


Figure 7. Reachability heat maps with total score per key (left) and score per finger correlated to the preferred finger (right).

b9, b13 and from y13 to y16). In that regard, the right part of the Figure 7 shows another heat map with the reachability score per finger correlated with the preference results, the scores presented are those of the preferred finger (previously reported in Figure 6) and can theoretically vary between 0 (the preferred finger of the key is also considered *unreachable* for all participants) and 66 (the preferred finger of the key is *easily reachable*, without effort for all participants, 3 x 22). For the three keys (b1, b4, b11) with a preference score shared by two fingers, only the best reachability score per finger is reported. A set of heat maps with reachability scores per finger is shown in Appendix B. The best reachability scores (green and orange areas) per finger reflect users' preference for using a particular finger over a key.

5.3.3. Dataset

The collected data from the questionnaire may be used for different kinds of cubic-shaped device, beyond the text entry interfaces; it is released for scientific use as supplemental online material. The raw data of the finger-to-key preferences and reachability represent 19,514 distinct entries, allowing changes to the weighting formula and further statistical analyzes.

6. Cubic Layout Design

This section, dedicated to the design of the cubic layout, integrates, and extends the analyzes and discussions of the results from both user studies. The cubic layout adopted by the keycube prototype is QWERTY based. The design choice is driven by a combination of the literature, the keycube design philosophy and the results of the two previous user studies.

Literature. There have been attempts to either improve or to get away from the traditional QWERTY layout, such as the Dvorak Simplified Keyboard (DSK) or alphabetical layouts (Klemmer, 1971; Norman & Fisher, 1982). However, some studies

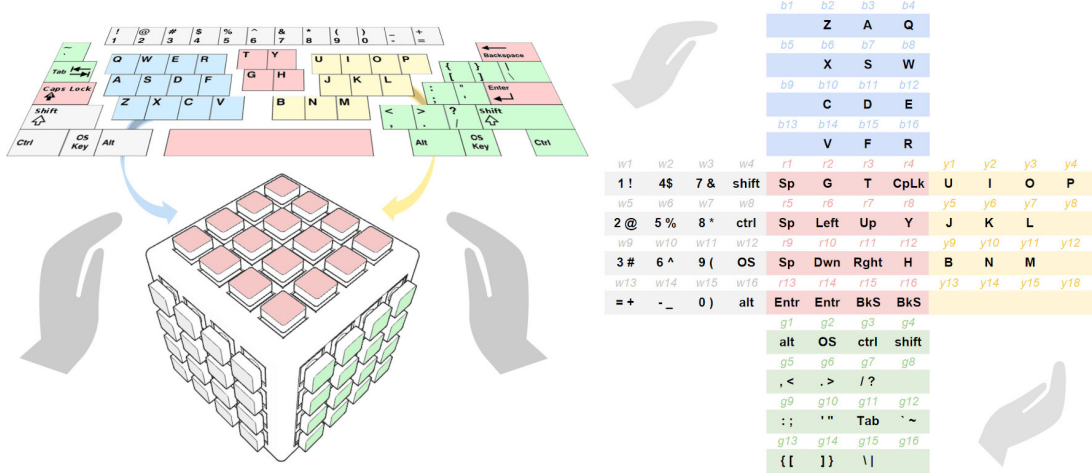


Figure 8. QWERTY layout cubic transformation.

point out that a change, in particular for the letters, is worthless due to minimal benefit (Buzing, 2003; Noyes, 1983b). Furthermore, sticking with the QWERTY layout while designing a new text entry interface has shown to be beneficial due to potential knowledge transfer (Matias et al., 1993; Scott et al., 2010; Wilson & Agrawala, 2006). Finally, some studies in economics (Kay, 2013) have suggested that the actual QWERTY layout entered a technological lock-in, even if shown inferior to alternatives, and it will be hard to move away, mostly due to “*technical interrelatedness, economies of scale, and quasi-irreversibility of investment*” (David, 1985).

Consistency. The design criteria behind the hardware part of the keycube were to be as close as possible to the characteristics of traditional physical keyboards (such as size, spacing and number of keys). Therefore, to stick with the objective of capitalizing on knowledge transfer between the two devices, we have adopted the same strategy regarding the cubic layout.

However, the cubic form factor of the device implies some adaptations. The three-dimensional transformation adopted and the whole cubic layout are shown in Figure 8. The aim was to keep as much as possible left-right hands relationship as well as fingers-to-keys associations from touch typing.

The external (blue and yellow) faces were reachable to all fingers except the thumbs and are only dedicated to the most used characters, the letters. The top face keys (red), intended to be pressed with the thumbs and which received high reachability score correlated with preference, supported very common special keys: Space, Enter, Backspace, the four directional arrows (for caret control while editing text) and 4 letters (T, Y, G, and H) in order to leave free the lowest lines of keys in the blue (b1, b5, b9, b13) and yellow (from y13 to y16) faces. Several criteria were used to justify this design deviation (with respect to finger-key relations since the thumbs are usually not used to press letter keys on traditional physical keyboards):

- The lowest key lines (rows) of the blue and the yellow faces are just as capable to receive the remaining central letters but they are close to the edges of the device in contact with the palms of the hands, thus, are likely to cause the accidental presses reported by 35.5% of the participants in the first study.
- The total reachability scores of the b9, b13, y13 and y14 keys (mean=79.5, SD=6.5) which can accept respectively the letters G, T, Y and H are slightly

lower by 15.2% than those of the actual chosen keys, r3, r3, r8, r12 (mean=93.8, SD=4.7). The difference is even more marked (from single to double) for the reachability scores per finger correlated with preference, meaning that the chosen keys (over the red face) are more reachable for the thumbs than the remaining free blue and yellow keys can be for the index and middle fingers.

Furthermore, our choice to design the cubic layout based on the QWERTY is paradoxically accentuated by a previous criticism of this layout concerning the lines of letters distribution regarding their frequency of use with the English language (Noyes, 1983b). It is stated that too much typing is carried out on the back row (the one starting with letters QWER), representing 52%, while the home row (the one starting with letters ASDF) represents only 32%, and 16% with the front row (the one starting with letters ZXCV). However, the three-dimensional change (from plane to cubic shape) of the layout, allows us to take advantage of this drawback while adopting the QWERTY because there is an actual increase of reachability between the three rows (Figure 7), starting from the front row (b2, b6, b10, b14, y9 to y12), the home (b3, b7, b11, b15, y5 to y8) and back row (b4, b8, b12, b16, y1 to y4) distributed over the two external faces (blue and yellow). Therefore, with a cubic shape, the most reachable row is the most used and the least reachable row is the least used (for the English language).

All other special characters of the cubic layout (Figure 8) currently disposed on the internal faces (white and green) for a better visibility are easily reachable by the thumbs but remain to be studied. The keys with numbers (placed on the white face) follow the order presented in a traditional numerical keypad, with the aim to ease their mental representation and decrease their retrieval time. Special characters dedicated to punctuation marks were kept on the right side (green face) following the same column order (left to right placement). However, they are in an inverse order of row placement (top to bottom) following a better reachability score on the top of the green face. Two special characters (Tabular and Tilde) usually dedicated to the left hand were placed on the right side. To preserve the multiple shortcuts and commands operated by combinations (e.g., Control-C), all modifier keys (such as Shift, Control or Alternate) are symmetrically doubled at the top (most reachable part) of the white and green faces.

7. User Study 3: Knowledge Transfer, Learning Curve and Performance

The goal of this comparative study was threefold: (1) upon initial exposure, verify if the keycube prototype and its cubic layout could benefit from knowledge transfer of the traditional keyboard, (2) bring out a learning curve of using the keycube, (3) measure the usability and workload induced by the device.

7.1. Study Design

A mixed experiment was designed (3 groups of 8 participants) with two independent variables: layout and session. Layout was between-subjects with 3 levels: QWERTY, AZERTY and Unknown (Dvorak-based). Session was within-subjects with 5 levels (session 1 to 5). The dependent variables were input speed, error rate and keystrokes per character for the whole experiment, as well as the subjective NASA Task Load Index (NASA-TLX) and System Usability Scale (SUS) only for the between-subjects factor (layout). Furthermore, to extend comparative evaluations, a sixth session was

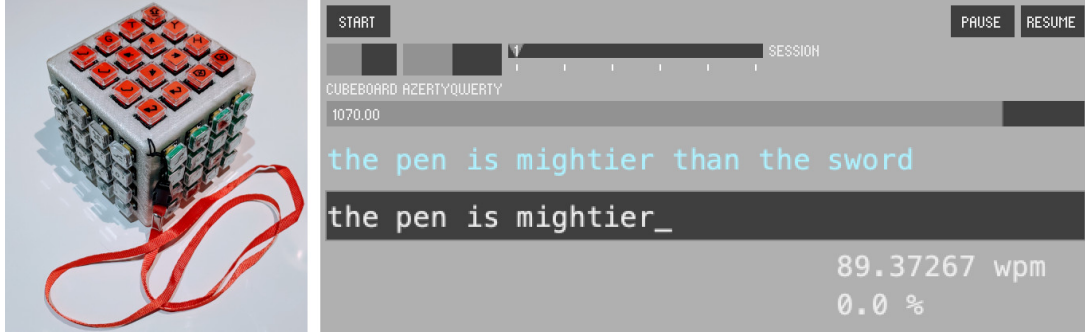


Figure 9. Third experiment hardware and software.

included with the traditional keyboard as a baseline condition.

7.2. *Participants*

A total of 24 participants (4 left-handed; 15 women; 19-36 years old, mean 28) have been recruited mostly from university campuses; 7 participated in the previous studies, four months separated this third study from the second, to avoid any experience bias. Participants used a traditional keyboard every day with either QWERTY (n=12, 50%) or AZERTY (n=12), and were ignorant of the DVORAK layout. To be sure that participants knew their respective layout, they had to type with a minimum input speed of 30 words per minute (wpm) with a traditional keyboard, still allowing a wide range of typing skills. Three of them reported to not watch the keys while typing (touch typing). Participants were required to read, write and understand the English language. All participants had given their written informed consent and received 35 CAD or 24 EUR as compensation.

7.3. *Materials*

Three new keycube prototypes were built and included minor updates based on results from previous studies. The touch screen was replaced by a 3D printed flat surface, decreasing the overall weight to 383 g (-11% less than the previous prototype). A neck (or wrist) strap was added to the structure for safety reasons and freedom of hands (Figure 9, left). Moreover, the strap placed at the level of an upper vertex (at the intersection of the red, white, and green faces) adds a tactile landmark which helps to orient the device without having to look at it. The keycubes were connected via Bluetooth to a laptop and the traditional QWERTY and AZERTY keyboards used for the text entry interface baseline were both Apple’s keyboards with low-profile keys.

For comparative purposes, the key covers with character inscriptions were disposed on each keycube following a different cubic layout. One was defined as the QWERTY-based cubic layout presented in the previous part (Figure 8). Undefined (empty) keys, such as y13 to y16, were covered with fully transparent key covers to avoid physical unbalance. The two other cubic layouts were based on AZERTY and Dvorak.

	b1	b2	b3	b4		b1	b2	b3	b4	
		W	Q	A			P	J	O	
	b5	b6	b7	b8			b5	b6	b7	b8
		X	S	Z			Y	Q	A	
	b9	b10	b11	b12			b9	b10	b11	b12
		C	D	E				F	K	U
	b13	b14	b15	b16			b13	b14	b15	b16
		V	F	R				G	X	E
w4	r1	r2	r3	r4	y1	y2	y3	y4		
shift	Sp	G	T	CpLk	U	I	O	P		
w8	r5	r6	r7	r8	y5	y6	y7	y8		
ctrl	Sp	Left	Up	Y	J	K	L	M		
w12	r9	r10	r11	r12	y9	y10	y11	y12		
OS	Sp	Dwn	Rght	H	B	N				
w16	r13	r14	r15	r16	y13	y14	y15	y18		
alt	Entr	Entr	BkS	BkS						
	g1	g2	g3	g4						
	alt	OS	ctrl	shift						

	b1	b2	b3	b4		b1	b2	b3	b4	
		P	J	O			P	J	O	
	b5	b6	b7	b8			b5	b6	b7	b8
		Y	Q	A			Y	Q	A	
	b9	b10	b11	b12			b9	b10	b11	b12
		F	K	U				F	K	U
	b13	b14	b15	b16			b13	b14	b15	b16
		G	X	E				G	X	E
w4	r1	r2	r3	r4	y1	y2	y3	y4		
shift	Sp	B	I	CpLk	H	T	N	S		
w8	r5	r6	r7	r8	y5	y6	y7	y8		
ctrl	Sp	Left	Up	D	W	V	Z			
w12	r9	r10	r11	r12	y9	y10	y11	y12		
OS	Sp	Dwn	Rght	M	C	R	L			
w16	r13	r14	r15	r16	y13	y14	y15	y18		
alt	Entr	Entr	BkS	BkS						
	g1	g2	g3	g4						
	alt	OS	ctrl	shift						

Figure 10. Two-dimensional unknown-based cubic layout.

7.3.1. AZERTY-based and Unknown-based Cubic Layouts

An adaptation of the QWERTY-based cubic layout was made to provide the AZERTY-based cubic layout, also supposedly familiar to participants and thus candidate to knowledge transfer. The AZERTY is a common layout in France and in regard to the letters, it is a slight variation from the QWERTY, therefore only the keys Q and A, W and Z, M and the key code y8 were switched (Figure 10, left).

On the other hand, the Unknown-based cubic layout was defined to be unknown from the participants, and thus avoiding potential benefits to previous knowledge from the participants. Instead of randomizing the assignment of characters, the cubic layout (shown in Figure 10, right) was based on the Dvorak Simplified Keyboard (DSK) layout optimized for the English language (Buzing, 2003). We applied the same strategies adopted to transform the QWERTY layout; however, the original home and top row have been swapped because the original DSK layout is optimized to use more often the home row with English language (which had a lower reachability score than the top row with a cubic shape).

Among the three cubic layouts, only the letter placement was changed because none of the numbers or special characters except Space, Enter and Backspace would be involved with the English phrases set of this study.

7.3.2. Software

A new software made shown in Figure 9 (right), presented easily memorable English sentences from (I. S. MacKenzie & Soukoreff, 2003) phrases set commonly encountered in text entry studies. The sentences are representative of the English language and do not include any numbers or special characters. The very few instances of uppercase characters were lowered. The software also displayed the remaining time of the experiment as well as the input speed and the error rate of the last sentence transcribed. All key events (press and release) were logged with timestamps.

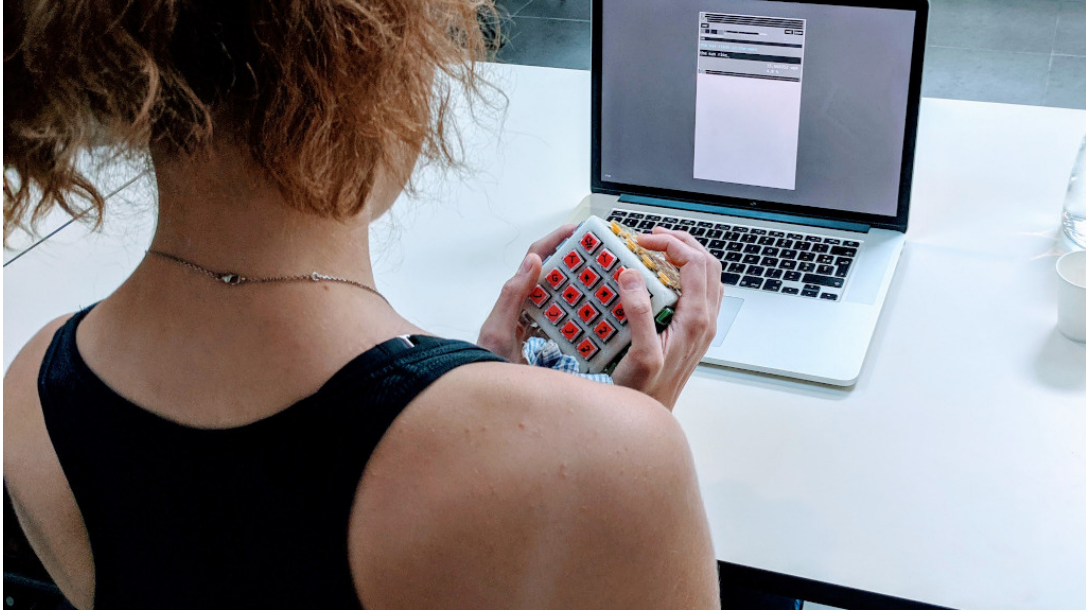


Figure 11. A participant, seated, using a keycube prototype in the third user study setup.

7.4. Procedure

After presenting the procedure of the study to the participants, we asked them to sign a consent form and to take a seat next to the laptop (Figure 11). Then, we asked them to fill a pre-study questionnaire on demographics and typing skills. The participants had to read and type a couple of random phrases from the set, to briefly check their current typing speed and understanding of the English language.

Finally, one of the 3 keycubes was assigned to them (representing 3 groups of 8 participants) depending on their traditional keyboard layout usage. Due to similarity of letters placement among the QWERTY and AZERTY layout, we did not expect a difference of traditional keyboard skills. Therefore, to the group of participants with the Unknown-based cubic layout, half of them usually used a QWERTY layout and the other half an AZERTY layout. The instructor demonstrated how to hold the device (diagonal position), but the participants were allowed to support it as they wished. The instructor only showed the needed keys for the experiment (letters, Space, Backspace and Enter keys), without giving any explanations of the layout. Participants were told to avoid as much as possible pressing unused keys (e.g., white and green keys) and were specifically instructed to “enter the text as quickly and accurately as possible” and started a warm-up round by typing the alphabet twice.

Then, the experiment was composed of 6 typing sessions of 20 minutes each, including the baseline interface. Short (SB) and long breaks (LB) were inserted between the sessions as such: S1 (session 1), SB, S2, LB, S3, SB, S4, LB, S5, SB, S6. Short break lasted around 1 minute and long break around 30 minutes (at least 20 minutes and at most 80 minutes). We included two long breaks because typing as quickly as possible for 20 minutes twice separated only by a short break is an exhausting activity. Participants were instructed to not use a traditional keyboard during the breaks but were allowed to use soft keyboards (for instance, with a smartphone). The first 5 sessions (S1 to S5) were performed with the keycube. The last session (S6) was performed with a traditional keyboard and participants had to type the alphabet twice in addition to

a couple of phrases as warm-up and to decrease potential bias due to the previous keycube use. For each participant, the whole experiment was conducted during the same half-day, and we assumed that one session (20 minutes) was enough for assessing the traditional keyboard baseline. Each session was run similarly and consisted of randomly selected English phrases to be typed by the participants. Straight after session 5 (keycube) and session 6 (traditional keyboard) participants assessed their subjective mental workload using the NASA-TLX (Hart & Staveland, 1988) followed by their satisfaction with the System Usability Scale (Brooke, 1996) questionnaire concerning their previous session only. In total the experiment, including the breaks, lasted around 3 hours and 50 minutes per participant.

In accordance with the practices and for the integrity of comparisons with literature, this procedure is inspired by previous user studies related to at least three other text input devices, a mini-QWERTY keyboard, a chord keyboard, and a pair of touch-sensitive gloves dedicated to mixed reality headsets (Clawson et al., 2005; Lyons, Plaisted, & Starner, 2004; Lyons, Starner, et al., 2004; Whitmire et al., 2017). In particular, we share with these studies: the duration of 20 minutes per session and the set of phrases. However, we condensed all 20-minute sessions into half-a-day, to be representative of a very intensive text entry activity, similar to those that are encountered in the context of office workspaces.

7.5. Results

During the sessions and across all participants, we recorded 153,450 key presses and 4,822 phrases (respectively 6,394 and 201 on average per participant) typed with the keycube. We removed all phrases with an error rate above 20% to exclude outliers (occurred 2.05%), they were caused by pressing the Enter key too soon.

7.5.1. Input Speed

The input speed (measured per sentence and then averaged) is calculated in Words per Minute (WPM) as

$$WPM = \frac{|T| - 1}{S} \times 60 \times \frac{1}{5}$$

where T is the transcribed text and S is the time in seconds calculated from the first key press to the last characters pressed. A word is defined as five characters including spaces.

The input speed per cubic layout is reported in Figure 12 in words per minute (left) and relative to the traditional keyboard baseline (right). A two-way analysis of variance (ANOVA) showed a significant effect both for the layout ($F_{2,105} = 83.761; p < .05; \omega^2 = 0.510$) and for the session ($F_{4,105} = 11.288; p < .05; \omega^2 = 0.127$), confirming a learning effect. The average text entry speeds with the cubic QWERTY, AZERTY, and Unknown layouts were respectively 12.1 (SD=3.0), 10.7 (SD=1.7) and 6.6 (SD=0.9) wpm for the first session, up to 18.7 (SD=4.8) 16.6 (SD=3.0) and 9.0 (SD=1.6) wpm at the last session, representing an increase of 54.8%, 55.2% and 35.2%. For the last session, the text entry speed averages for the QWERTY-based and AZERTY-based layouts are nearly twice (respectively 107% and 84%) as much as the average text entry speed of the Unknown-based layout. A post-hoc analysis using Bonferroni pairwise t-test was performed to determine no significant effects on input speed between the

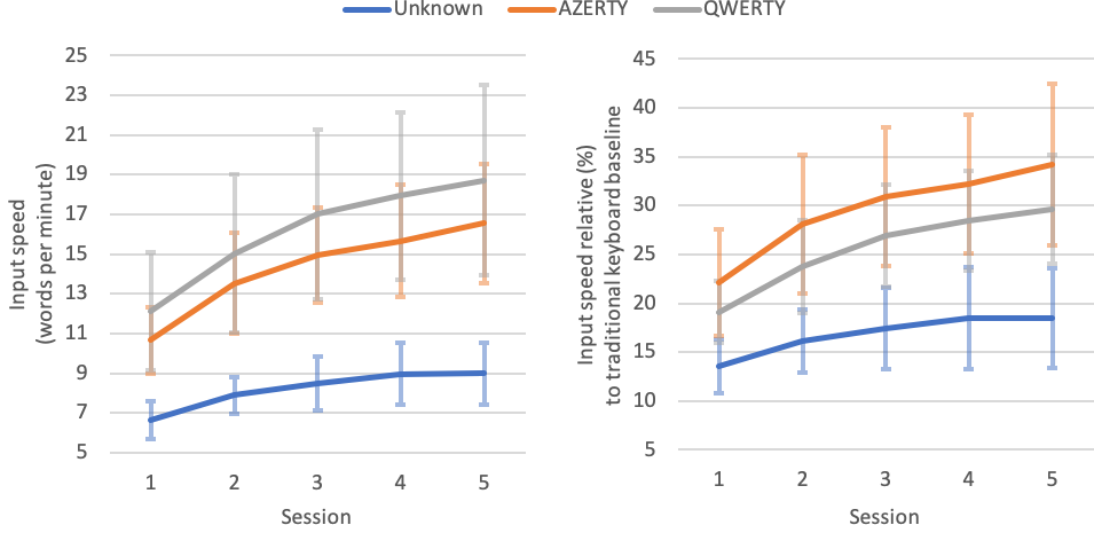


Figure 12. Average participants input speed per session for each cubic layout (Unknown, AZERTY, QWERTY).

QWERTY-based and AZERTY-based cubic layouts for the first ($p = .255$) and last session ($p = .298$).

We logged every key press from unused keys (e.g., white and green faces of the keycube). Participants reported pressing unintentionally a few keys mostly with their hand palms. For the last session with the keycube (S5), this issue happened in 2.53% among all participants and may have negative consequences on the input speed (decreased) by forcing them to be aware and to use the device more sensitively.

Besides, some participants ($n=14$) reported out loud an issue regarding repeating characters while pressing a key only once (caused by switch bouncing). We analyzed the logs and detected this issue with the inter-key intervals (IKI), delay between two key presses, which includes the time required for a user to search and find a key. Repeating characters caused by hardware key switch sensitivity had a low inter-key interval (usually below 5 milliseconds) and were considered as outliers. With a threshold of 50 milliseconds, this issue represented only 0.75% of the total key presses during the fifth session (S5).

7.5.2. Error Rate (uncorrected) and Keystrokes Per Character (KSPC)

We used the updated version of MacKenzie and Soukoreff error rate (I. S. MacKenzie & Soukoreff, 2002a) based on the Levenshtein string distance and is calculated as

$$ER = \frac{MSD(A, B)}{\bar{S}_A} \times 100\%$$

where $MSD(A, B)$ is the minimum string distance between the presented text A and the transcribed text B, and $\bar{S}_A$ is the mean size of alignments.

The keystrokes per character (KSPC) is the ratio of the input stream length (which includes Backspace) to the transcribed text length. As a traditional physical keyboard, the keycube prototype is considered unambiguous for the phrases set used in this study because each letter has a single dedicated key, thus, the best KSPC is 1.0 and increases when an error is corrected.

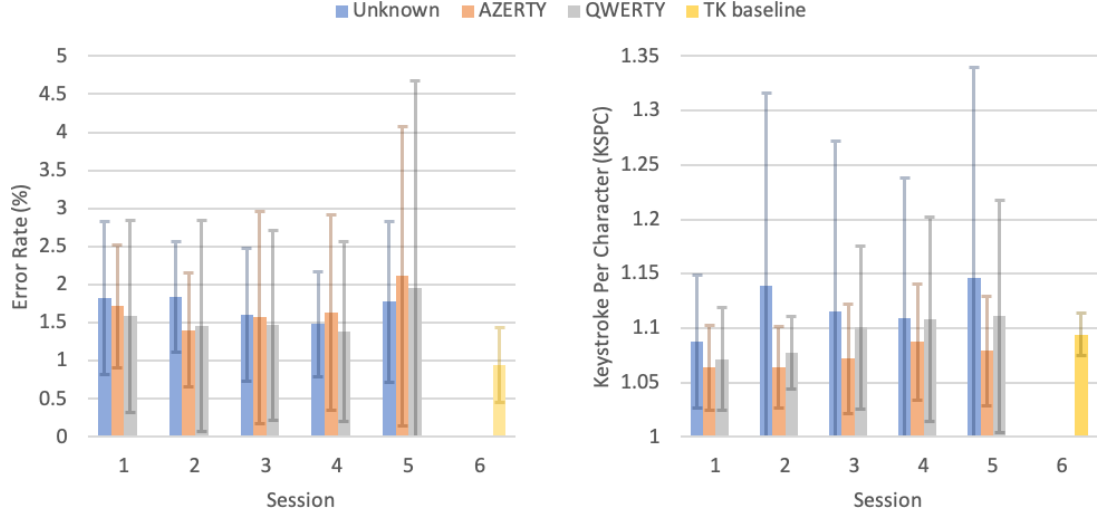


Figure 13. Error Rate (left) and Keystrokes Per Character (right) per session for each layout including comparison with the traditional keyboard baseline.

The error rate and KSPC per layout across all sessions, including the traditional keyboard (TK) baseline are presented in Figure 15. Two-way ANOVA (among cubic layouts only) showed that neither layout nor session had a significant effect on the error rate (with respectively $p=0.883$ and 0.768) and on the KSPC (with $p=0.128$ and 0.772). The average error rate for all sessions and layouts is 1.66% (SD=1.26), while the KSPC is 1.09 (SD=0.10). All participants across each group seemed to have adopted the same behavior of correcting most of the errors (thus, low error rate) and avoiding as much as possible pressing the wrong keys (thus, low KSPC).

7.5.3. Workload

The perceived workload was assessed by the NASA-TLX, consisting mostly of six dimensions (mental demand, physical demand, temporal demand, performance, effort, and frustration) varying on a 21-point scale presented in a questionnaire. A final index mixing the dimensions is given following a formula presented in the original paper (Hart & Staveland, 1988). A lower index is preferred and indicates a lower level of workload. All measures per dimension and the raw index (the sum of all dimensions normalized) for each cubic layout and traditional keyboard are shown in Figure 14. The workload raw index for the Unknown-based cubic layout (mean=59.6, SD=12.6) is higher (worse) than the raw index of the QWERTY-based (mean=52.9, SD=13.24) and AZERTY-based (mean=53.1, SD=10.7) cubic layouts. This difference is particularly marked and caused by the performance and frustration dimensions. Due to intensive familiarity, the traditional keyboard (mean=35.8, SD=13.8) was expected to receive a less demanding task workload. For the participants having the traditional keyboards with the QWERTY layout, the raw index was 31.9 (SD=11.7), and 39.6 (SD=15.2) with the AZERTY layout.

7.5.4. Usability

The usability is assessed with the System Usability Scale (SUS) method. The SUS gives a score calculated on a 0-100 scale following a formula based on ten 1-5 Likert-scale questions (Brooke, 1996). A higher score on the scale is preferred and indicates



Figure 14. NASA Task Load Index per dimension (left) and raw score (right) for each cubic layout and traditional keyboard baseline.

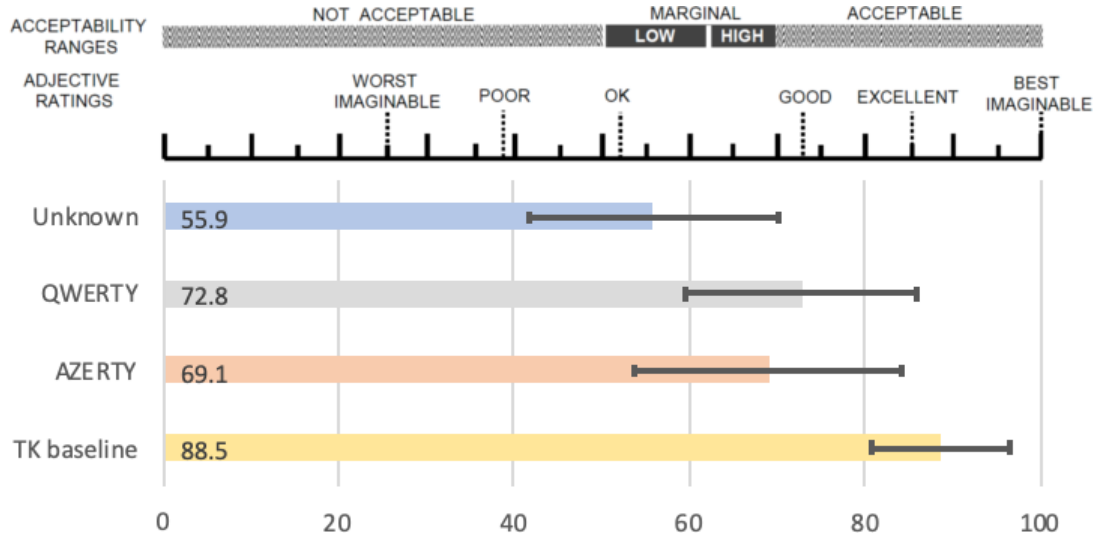


Figure 15. SUS score per layout with the acceptability ranges and adjective ratings from Bangor et al. (2009)

a higher level of usability. Our results for each cubic layout and traditional keyboard are reported in Figure 15. The QWERTY-based, AZERTY-based and Unknown-based cubic layouts received respectively a mean SUS score of 72.8 (SD=13.3), 69.1 (SD=15.2) and 55.9 (SD=14.1). Both QWERTY-based and AZERTY-based layouts entered the “Acceptable” ranges and are respectively 30% and 24% higher than the Unknown-based cubic layout. The mean SUS score of the traditional keyboard (TK) baseline reached 88.5 (SD=7.9). For the participants having the traditional keyboards with the QWERTY layout, the score was 89.6 (SD=8.4), and 87.5 (SD=7.6) with the AZERTY layout. These results are consistent with previous research showing that a generic device (such as the traditional keyboard) should be scoring around 15-16% more than a new prototype due to an extensive familiarity (Kortum & Bangor, 2013).

7.6. Discussion

All the results were consistent, both QWERTY and AZERTY cubic layouts are similar with more advantageous results than the Unknown-based cubic layout, meaning that

there was indeed a transfer of knowledge, in particular to the input speed (from single to double) and the usability (30% higher). A knowledge transfer also supported by open comments from 37.5% of participants (6 out of 16, as it was expected only for groups with QWERTY or AZERTY-based cubic layouts) who mentioned "*it is easier than what I thought*", comments particularly encouraging as they were spontaneous, without reflection and out of the questionnaire.

Keycube users with the QWERTY-based cubic layout achieved 19 wpm within five 20-minute sessions of practice in a single half-day, accounting for 30% of their input speed with a traditional keyboard, for a comparable error rate. With the same criteria, the best input speed performance was 25.7 wpm (with a 1.07% error rate) and the best relative input speed was 45.7% from another participant. These results are even more encouraging when put into perspective with related works using the same set of sentences in 20-minute sessions spread over multiple days. Twiddler chord keyboard users reached 13 wpm after the 5th session to finally achieve the same input speed as the keycube later at the 10th session (Lyons, Starner, et al., 2004). Moreover, keycube users were four times quicker than Twiddler users to reach the relative input speed to traditional keyboard of 30%, and thus addressing the main weakness of chord keyboard: long learning requirements. With DigiTouch (Whitmire et al., 2017), a touch-sensitive pair of gloves capitalizing on the QWERTY layout, users reached around 12.5 wpm after the 5th and up to 16.0 wpm for the 10th (an input speed already reached by keycube users at the 3rd session of the same half-day).

With the RearType (Scott et al., 2010), a keyboard behind a tablet also capitalizing on the transfer of knowledge from QWERTY layout, users reached 15 WPM (and 21% of their input speed with traditional keyboard) after entering the same text for 60 minutes. The results of the keycube are also comparable to those of the Half-QWERTY (a traditional keyboard reduced by half), for an equivalent practice time, the participants reached 18.3 wpm (Matias et al., 1993). However, both RearType and Half-QWERTY interfaces were advantageously tested with traditional keyboard experts and used tutorial techniques to increase skill acquisition and thus their results.

The input speed and error rate are not the only important characteristics of a text entry interface, several participants (n=5) reported that the small form factor of the device allowed them to adopt more relaxing positions than with a traditional keyboard, for example, with the back against the chair and the arms along the body, resting on the thighs. In this regard, the keycube, beyond bringing mobility during its use, could also follow the unusual history of the split keyboard design adoption (Rempel, 2008) that put improving health above productivity. Even though the participants were seated in front of a laptop computer, most of them (n=18, 75%) handled the keycube without using the table throughout the whole experiment, holding the device with the forearms bent as it was the case during the first user study, thus keeping the mobility advantage. The other participants used the table a few times as a support for their elbows, forearms or even directly to put the device down.

No strategy has been adopted to limit users' gaze at the device, yet, 6 participants openly reported that they already started between the 1st and 3rd session to partially enter text eye-free, without having to focus their gaze on the corresponding key.

Finally, the input speed decrease for the last session (S5) of the Unknown-based cubic layout as well as comments from 8 participants suggest that the experimentation was very demanding, including for groups with the QWERTY and AZERTY cubic layouts. Despite the short and long breaks, the instruction to enter text as quickly as possible intensively for 1 hour 40 minutes (within 3 hours and 20 minutes of the same half-day) remains a (purposely) exhausting task. In this condition, not surprisingly,

the device itself may have caused some muscle fatigue, mentioned by 9 participants but for different locations of the body, little fingers (n=2), hands (n=2), wrists (n=2), back and chest (n=3), most often the reasons given were the weight (n=3) or size (n=9) of the cubic device considered slightly too high. One participant who commented: "Physically it is very different, but surprisingly I feel quite comfortable" also mentioned that the button actuation force could be reduced. In this regard, adopting different commercial-grade quality of material could reduce the total weight of the device and slightly decrease the size without compromising on the design philosophy and criteria of the keycube. Moreover, other strategies could be adopted, such as implementing keys with different actuation forces depending on their location or their frequency of use, for instance, by only increasing the actuation force of the buttons close to areas subject to accidental presses (near the edges in contact with the palms of the hand) and by reducing that of the buttons that are deliberately regularly pressed.

8. User Study 4: Augmented Reality Context and Fingers Usage

After investigating the potential and performance progression of the cubic shaped text input device in a seated context, we conducted another study to explore and compare the performance of the keycube with an augmented reality headset in mobile context, showing its versatility of the postures adopted by its users, in order to spatially exploit the virtual workspace. Moreover, in a second part, we measured and compared the use of fingers to press the keys with a cubic shaped device (keycube) to detect eventual advantages compared to two-thumbs-only text entry.

The study was within-subjects with one independent variable: the text entry interface for the augmented reality headset, which was either using the keycube, or two baselines : a virtual keyboard on a smartphone (VKS) or a head-then-hand gesture keyboard (HHGK). The dependent variables were input speed, error rate, keystrokes per character, usability, and workload. For further comparative reasons, we also measured the text entry performance of the user with a traditional keyboard.

We could not find a text entry interface sharing the exact same usage characteristics aimed by the keycube (i.e., device usable in various postures with numerous tactile switches based on a QWERTY layout directly accessible from multiple fingers also allowing combination shortcut commands and eye-free possibility), therefore, the baseline interfaces were mostly integrated to validate the formality of our user study procedure since their performances were already known from the literature. In this regard, participants are expected to receive better performances (from all criteria) with the keycube than the Head-then-Hand Gesture Keyboard (HHGK), considered truly novices for both interfaces, but the opposite is expected for the Virtual Keyboard on Smartphone (VKS), especially since they are already experts in the latter.

8.1. Participants

Nine volunteers have been recruited from Le Mans University campus (5 women, 4 men; 24-35 years old, mean 28), none of them participated in the previous study, but had the same requirement. Every participant used a traditional keyboard with an AZERTY layout on a daily basis. Every participant used a smartphone many times per day, 5 of them with autocorrect text entry, 7 of them with word suggestions and none of them used the word-gesture feature. Every participant received 10 EUR as compensation for their participation.

8.2. Materials

We used a keycube with the AZERTY-based cubic layout, an augmented reality headset (HoloLens from Microsoft, 1st generation), a smartphone (Google Pixel) and a laptop with the AZERTY traditional keyboard.

8.2.1. Software

The application from the previous study has been ported to Unity, now compatible with the augmented reality headset and the laptop. The application allowed participant to enter text in 4 different ways, with a traditional keyboard (only on the laptop) or either with the three interfaces dedicated to the augmented reality headset (Figure 16):

- **Keycube.** To simplify the implementation process, the keycube was connected to the smartphone over Bluetooth. The smartphone was solely considered as a gateway and sent the data back to the augmented reality headset over Wi-Fi.
- **HHGK** for Head-then-Hand Gesture Keyboard. The head movement is used for pointing and selecting keys on a virtual keyboard present in front of the user's field of view, then, an in-air hand gesture must be performed to enter the current selected character. Despite being similar to the text entry interface already implemented by default with the Microsoft HoloLens (1st gen.), we developed our own with minor changes to include an AZERTY layout and to remove unused special characters and numbers. This interface is still considered as a baseline to our study with known performances in the literature (Derby, Rarick, & Chapparro, 2019).
- **VKS** for Virtual Keyboard on Smartphone. A mobile application was developed to connect the smartphone to the augmented reality headset over Wi-Fi. For fairness with the two other interfaces, the intelligent features were deactivated. However, even if *autocorrect* has been shown to be positively correlated with input speed (Palin et al., 2019), in the same study, and another (Quinn & Zhai, 2016), *suggestion* has been negatively correlated to input speed. To our knowledge, text input performance from the keyboard on smartphones has never been studied while simply being coupled with an augmented reality headset. Yet, this interface could also be interpreted as a baseline since the device and the hands remain fully visible, we hardly expect different text entry performance than using the mobile phone alone, already known (Palin et al., 2019).

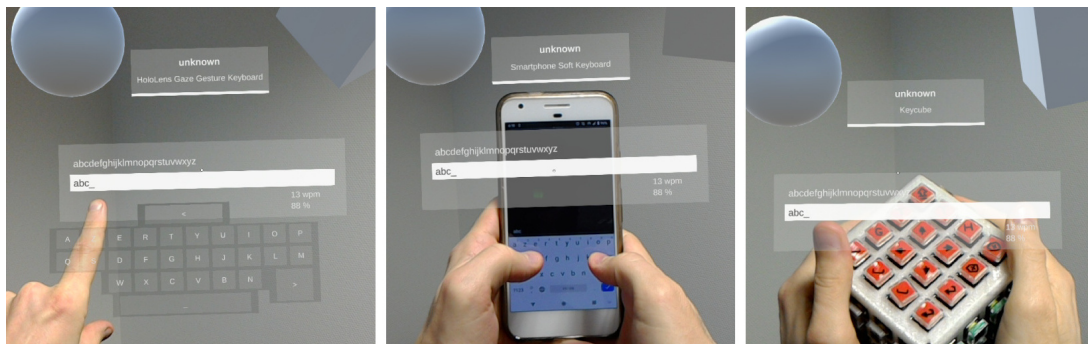


Figure 16. Visual interfaces from the application running on the augmented reality headset, from left to right: HHGK (Head-then-Hand Gesture Keyboard), VKS (Virtual Keyboard on Smartphone) and keycube.

8.3. Procedure

After individually presenting the procedure of the study to the participants, we asked them to sign a consent form and fill a pre-study questionnaire on demographics and typing skills. In the first part of the experiment, the participant performed a 5-minute text entry session seated with a traditional keyboard. Then, following the same instructions as the previous study, the participants were invited to perform three 20-minute text entry session, one for each interface, with a 5-minute pause between them. The interface order was counterbalanced following the Latin Square model. During the text entry sessions, participants were allowed to move freely, as long as they stayed standing (Figure 17). After each text entry session, the participants were asked to fill the NASA-TLX and SUS questionnaires. In a second part of the experiment, participants were asked to enter 13 sentences twice while seated, once with a keycube and once with a traditional keyboard, in both cases the hands and the device were recorded with two cameras for post-experiment analysis. The first sentence was a pangram used for training, the others 12 came from the same set used during the studies. In total, the experiment lasted around 120 minutes per participant.



Figure 17. Experiment setup, a participant standing using each of the interfaces with the augmented reality headset, from left to right: HHGK (Head-then-Hand Gesture Keyboard), VKS (Virtual Keyboard on Smartphone) and keycube.

8.4. Results

The results come from the analysis of the logs, questionnaire answers, video records and observation from the instructor. In total, 1353 sentences have been transcribed (481 with the keycube), 3.3% of them were excluded because they were outliers with an error rate above 20% caused by premature validation.

The input speed, error rates and keystrokes per character formulas, as well as usability and workload assessment methods are the same as the previous user study (section 7.5).

8.4.1. Input speed

The absolute (in word per minute, wpm) and relative (to the traditional keyboard) input speed for each interface are presented in Figure 18. Within one 20-minute session, the users are twice as fast with the keycube as with the Head-Then-Hand gesture keyboard (HHGK) baseline, respectively 10.0 wpm (SD=2.0) and 5.5 wpm (SD=1.5), representing 22.9% and 12.5% of their speed with a traditional keyboard. The input speed from virtual keyboard on smartphone (VKS) reached 34.0 wpm (73.8% from traditional keyboard), an expected result due to extensive familiarity (several years) with the interface. An analysis of variance (ANOVA) showed that the interface had a significant effect on the input speed ($F_{2,24} = 32.627; p < 0.05; \omega^2 = 0.717$). A *post-hoc* analysis using Bonferroni pairwise t-tests was performed to determine significant effects between each interface. Input speed results from the HHGK (baseline) are consistent with a previous study from the literature, mean=5.4, SD=0.9 (Derby et al., 2019).

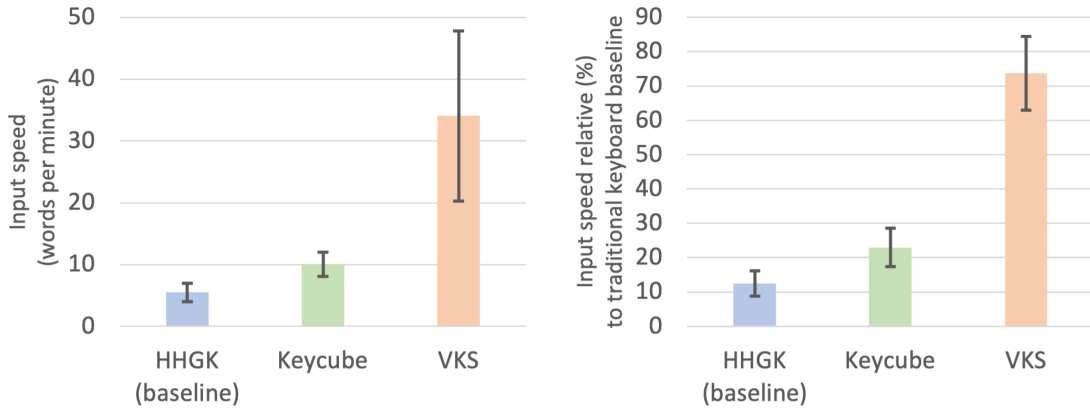


Figure 18. Absolute input speed (left) and relative to traditional keyboard (right) for each interface.

8.4.2. Error rate and keystroke per character (KSPC)

Since intelligent text entry features from the smartphone were deactivated, for all interfaces, the best KSPC were 1.0. The error rates and KSPC for each interface (including the traditional keyboard) are presented in Figure 19. An ANOVA (for all interfaces, including the traditional keyboard baseline) revealed that the interface had no significant effects on the error rate ($p = 0.186$), nor on the KSPC ($p = 0.209$).

8.4.3. Workload

The workload (from the NASA-TLX method) is presented per dimension and with the raw score for each interface in Figure 20. The workload raw score for the baseline interface HHGK (mean=66.8, SD=17.3) is higher (respectively at 45% and 58%) than the interfaces keycube (mean=46.0, SD=11.3) and virtual keyboard on smartphone (mean=42.1; SD=19.3). The slight difference between those two interfaces (keycube and VKS) mostly comes from the mental demand and effort dimension, higher for the former and from the frustration dimension, in this case, higher for the virtual keyboard on smartphone.

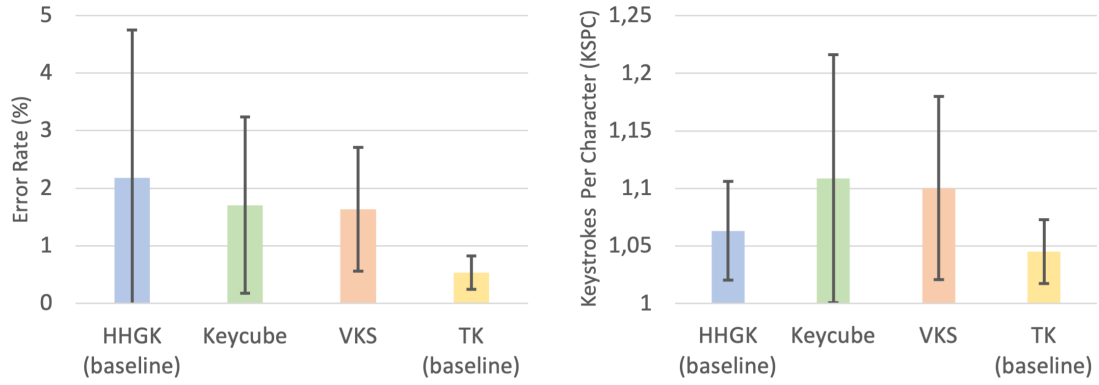


Figure 19. Error rates (left) and KSPC (right) for each interface.

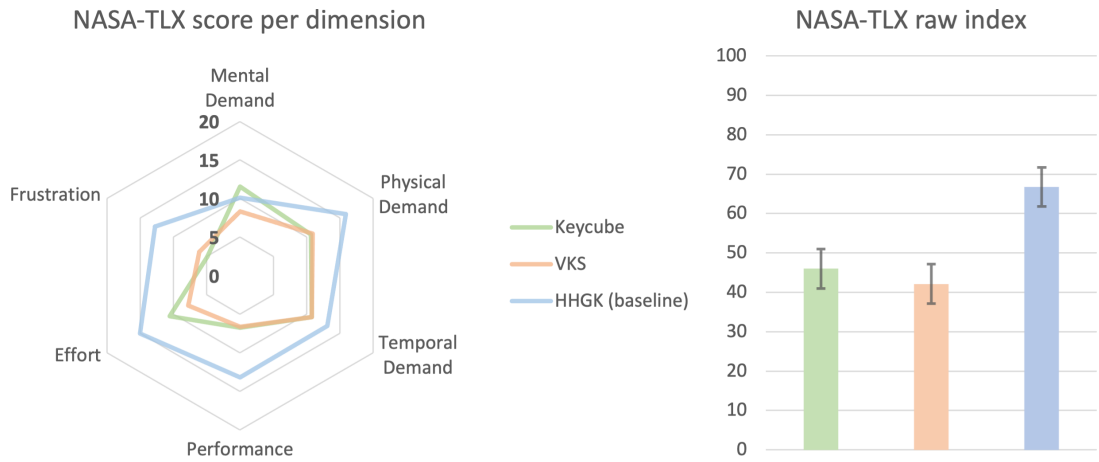


Figure 20. Workload (from the NASA-TLX method) score per dimension (left) and raw score (right) for each interface.

8.4.4. Usability

The usability score (from the SUS method) for each interface is presented in Figure 21. The usability score from the interface baseline HHGK (mean=45.8, SD=12.3) is considered not acceptable and much lower (respectively at 33% and 38%) than the interfaces keycube (mean=68.9, SD=12.2) and VKS (mean=73.9, SD=19.0). Even from a different group of participants, the subjective result of the HHGK (baseline) is also consistent with a previous study, mean=45.3, SD=13.9 (Derby et al., 2019).

8.4.5. Fingers usage

The results come from the analysis of videos, by bringing out the finger and the corresponding key for each recorded press.

The fingers usage rates for the keycube (after a 20-minute session) is presented in Figure 22. On the left part, the usage rate is averaged among all participants. However, as with a traditional keyboard, the cubic shaped device offers freedom in the choice of finger usage to press a key, for instance, some participants could mostly use the middle fingers while others the index fingers. Thus, on the right part, the figure presents the actual usage rates of the most used finger (regardless of the finger concerned) to the least used finger among all participants.

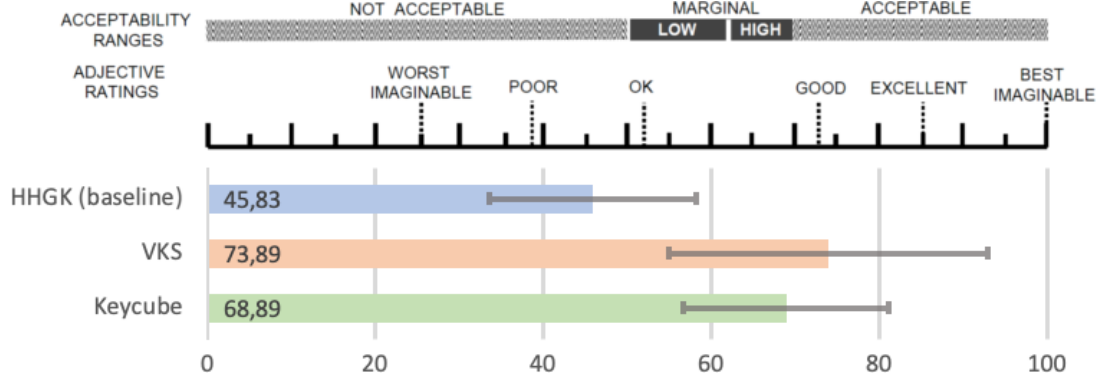


Figure 21. Usability score (from the SUS method) for each interface with the acceptability ranges and adjective ratings by Bangor et al. (2009).

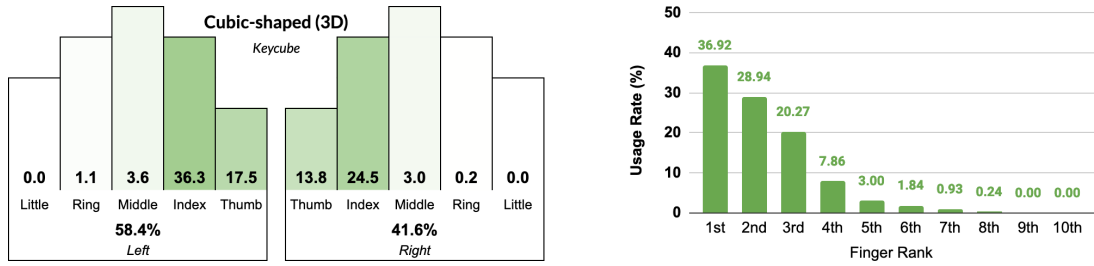


Figure 22. Usage rate per finger (left), and load ranking, regardless of the finger (right).

Eight fingers were used (all but the little fingers), yet, 92.1% of all key presses were done by four fingers, the index and thumb from both hands. The most used finger is the left index with 36.3%. Regardless of the finger, the actual highest load is similar at 36.9%. The lower the load, the better it is (e.g., when the key presses are shared equally among all fingers). This result shows benefit over two-thumb text entry (mostly used with smartphones) where at best, the two thumbs are equally used and both fingers have an already higher individual load, 50% each. If the thumbs are not perfectly equally used (most likely and worst scenario), one of them has an individual load even higher than 50%, worsening the repetition leading to muscular stress.

Although the form factor of the keycube encourages typing with multiple fingers, we noticed that participants were still able to press keys using a single finger, adopting hunt-and-peck strategy, often during the first few minutes, while supporting the device with the other hand to make visible and reachable simultaneously the three faces dedicated to letters (Figure 23).

Comparative finger usage analysis between the cubic (keycube) and plane (traditional keyboard) shapes is given in the Appendix C.

8.5. Discussion

In the first part of this user study, we showed that users could handle the cubic-shaped device standing up with an augmented reality headset while keeping the same text-entry performance to that of the previous user study while sitting in front of a laptop (section 7). An ANOVA between the results of the first session (S1) from the AZERTY group of the previous study and from the keycube session of this study showed that

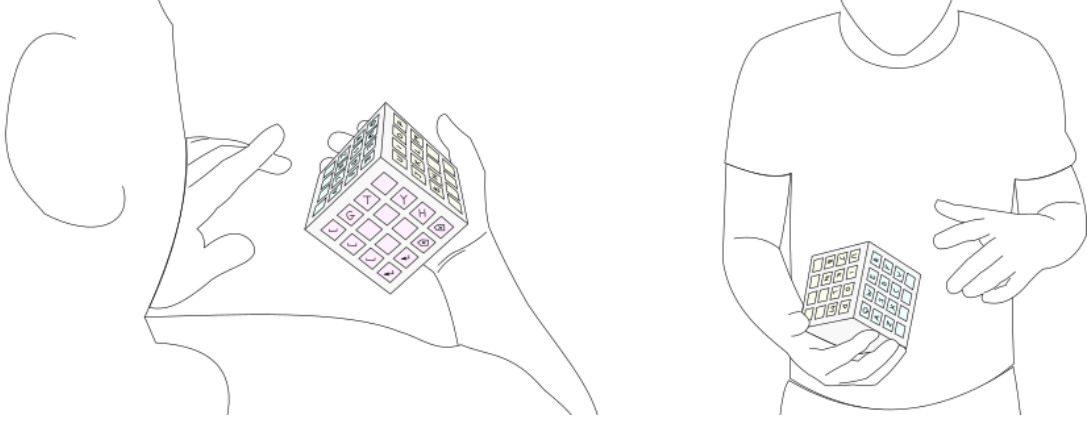


Figure 23. Three faces (blue, red and yellow) simultaneously visible and reachable by a single finger.

there was no significant difference in the input speed ($p=0.477$), error rate ($p=0.981$) and KSPC ($p=0.283$). Moreover, Figure 24 presents a more detailed intra-session with a very similar evolution of the input speed, error rate and KSPC. The sub-sessions are created by separating the transcribed sentences from the compared sessions in chronological order into three groups of equal size.

We did not measure the workload and usability of the traditional keyboard from the participants of this experiment, we cannot verify the consistency with the results from the previous user study, especially since the assessment concerned the last session of the keycube (S5) and not the first one (S1). Nevertheless, the results from both user studies are still coherent, for this experiment, there were fewer sessions than the previous one (1 against 5 with the keycube), therefore it was slightly less tiring, and the workload is 13.4% lower. The usability scores from both user studies are similar, with chronologically respectively 69.1 (SD=15.2) and 68.9 (SD=12.2).

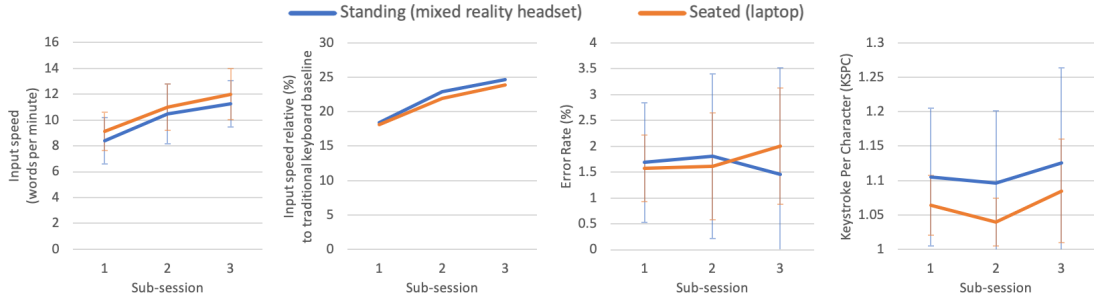


Figure 24. Comparison of the input speed (absolute and relative to traditional keyboard), error rate and KSPC for the keycube between sitting (with laptop) and standing (with the augmented reality headset) conditions.

Two baseline interfaces for augmented reality headset were integrated to the procedure, one is based on head movement followed by hand gesture and received the lowest overall performances for each criterion. The other interface is based on the virtual keyboard on a smartphone; thus we expected better performances due to extensive familiarity from the participants (years of experience). However, the KSPC, and more surprisingly, the workload and usability score were just similar to those of the keycube. Few reasons might explain this particular disappointing result of the smartphone-based solution. First, there is little to no room for improvement, the text-

entry performance such as input speed was not increasing during the session, users already reached their maximum from the beginning since they knew how to use a smartphone, on the contrary with the keycube, their performance kept increasing as reported by the Figure 24 and from participants comments "*Ah it's not bad, I am starting to learn.*" Second, typing as quickly as possible on only 20 minutes already shows the limit of smartphone keyboard for intensive text entry, one participant complained "*my arm hurts*" and another "*my neck is starting to hurt because I lower my head.*" Third, related to the previous point, the limited feedback (no activation force) and small size of the smartphone touchscreen make it impossible or hard to type without having to watch the screen of the device, breaking the immersion of the augmented reality experience. However, with the keys of the cubic shaped text entry device, users could already start typing a few characters without watching the device and at least foresee a use of the device fully eye-free.

In the second part of the experiment, a finger usage analysis showed that the keycube was used with up to 8 fingers and the highest finger load was 36.9%, already below the best scenario from the two-thumb text entry encountered with the keyboard on smartphones. This result is particularly interesting since the participants were just plain novices, but as shown in the next section, the quick expertise progression of the keycube, in addition to greatly improve the performance, keep reducing the highest finger load by sharing it among more fingers, in a way, similar to a traditional keyboard.

9. Case Study: Longitudinal

The aim of this case study is to give a glimpse and a possibility to the performance from one participant over a month period with a cubic-shaped text entry device.

9.1. Participant

The participant used the traditional keyboard everyday mostly without having to look at the keys. The participant is one of the keycube authors with a limited experience of the device when performing demonstrations as well as functional tests (either hardware or software), giving an initial advantage over completely new users. However, prior to this study, he had not performed any text entry session with the device.

9.2. Material

A black and white keycube configured with an AZERTY layout, but without any inscription (Figure 25) was used for most of the study. This keycube was connected to a laptop through a USB cable in order to reduce its components and the total weight to 324 gr. Another software was developed with Processing, similar to those of the previous studies but instead of presenting random sentences for 20 minutes, it presented 20 sentences from another commonly used set of 200 easily memorable English sentences, including this time: numbers, punctuation and upper cases (Vertanen & Kristensson, 2011). At the end of this case study, the colored and wireless keycube with inscriptions of the AZERTY-based cubic layout as well as the software dedicated to the previous studies were used.

9.3. Procedure

The participant performed 40 text entry sessions with the neutral keycube (black and white) limited to 20 sentences each, then two final 20-minute sessions with the colored AZERTY-based cubic layout keycube, respectively under the same conditions as the previous user studies 3 and 4. The sessions were spread over 20 days, thus, twice a day (in the middle and at the end). The last two 20-minute sessions were done on separate days. Thus, unlike the user study 3, the learning strategy followed a distributed practice, based on the psychological *spacing effect*, stating that learning sessions are more effective when distributed over time rather than massed, a phenomenon confirmed for typing skills (Baddeley & Longman, 1978). Between each session, the participant intensively used a traditional keyboard. Moreover, minor changes in the conditions have been conducted during this case study:

- (1) The first 20 sessions (half) were performed with the viewer on screen (presented in part 3.3), then without any visual aid for all subsequent sessions. The visual aid can be considered beneficial in the early stages of learning psychomotor skills (Fleishman & Rich, 1963), such as entering text with a keycube.
- (2) After the 30th session, we updated the software running on the keycube micro-controller to implement an anti-bounce system to reduce the problem of character repetitions reported in previous user studies. To avoid key bouncing (transition from one state to another), we included a delay (threshold to be exceeded) before considering changes in the key state, thus ignoring bounces made before that time limit. An analysis of the time differences of all the repeated characters (including those desired based on the vocabulary) from the previous sessions revealed outliers with a significant amount (31.5%) of repetition carried out in less than 5 milliseconds. We set the threshold at 25 milliseconds for safety, still keeping it relatively low to not slow down the pace of text entry.
- (3) To observe skill retention, a 5-day break was taken between sessions 30 and 31, as well as between the two 20-minute sessions.

To include comparisons, the participant also entered text with baseline interfaces: TK (Traditional Keyboard) and VKS (Virtual Keyboard on Smartphone, without any intelligent text entry feature). Once with the whole set of sentences, but also for two 20-minute sessions (one for each interface) in the same condition as the previous studies. Lastly, the participant followed the same procedure as the previous user study to record the finger usage.

9.4. Results

These quantitative results should be interpreted with some caution, they are by no means an indication of typical performances but represent nonetheless what could be achievable by some users. The results come from the logs and videos only. A total of 1092 sentences were transcribed with the keycube, those with an error rate greater than 20% caused by premature validation were excluded, representing 2.28% of the total.

Evolution of input speed (absolute and relative to TK), error rate (ER) and keystrokes per character (KSPC) for the first 40 sessions is shown in Figure 25. Input speed gradually increased from 13.6 to 44.7 wpm (session 38), representing an increase of 329% for a total practice time of 175 minutes (going from only 9 to 3 minutes of practice per session), just less than 3 hours spread over 20 days. The input speed rela-

tive to the TK started at 16.6% and reached 54.7%. At the fifth session, the participant achieved a speed of 20.9 wpm (slightly above the average from participants of the user study 3), and 36.3 wpm within approximately 100 cumulative minutes of practice, achieved during session 18. The error rate increased until session 12, from 0.29% to 5.85%, then stabilized, averaging at 5.44% over all of the subsequent sessions. At the same time, the number of keystrokes per character decreased continuously between the first and the last session, meaning that the user corrected less and less of his errors and they were less and less frequent. With the same set of sentences, the participant’s input speed in words per minute (wpm) and error rate (ER) with TK and with VKS was respectively 81.7 wpm, ER = 0.1% and 43.7 wpm, ER = 2.1%.

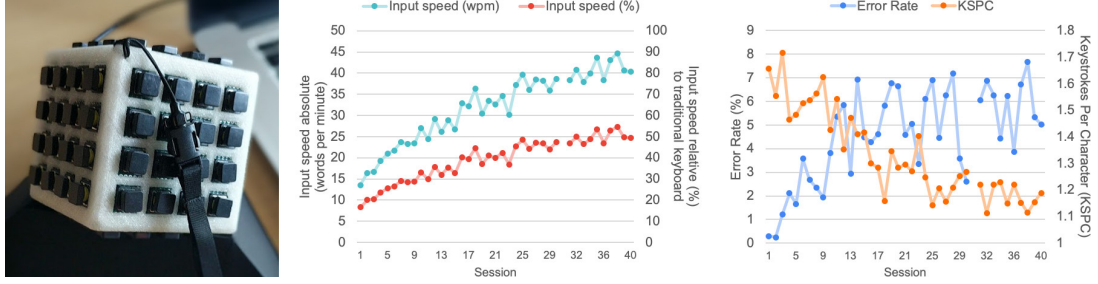


Figure 25. Black and white neutral keycube prototype (left), input speed (central), error rate and KSPC (right) for each session.

During the first 20-minute session with a keycube and conditions similar to the user study 3 (seated, in front of a laptop), the participant’s input speed was 48.4 wpm and ER = 5.4%, while the performance with the baseline interface TK was 86.8 wpm, ER = 0.7%. During the second 20-minute session, with condition similar to the user study 4 (standing, with an augmented reality headset), the participant typed 150 sentence, at an average of 55.1 wpm, ER = 6.7%, while the performance of the baseline VKS was 51.0 wpm and ER = 2.8%. Thus, during this last session, the input speed with the keycube relative to TK and to VKS reached respectively 63.5% and 108.0%.

Moreover, similar to the traditional keyboard, but unlike the virtual keyboard on a smartphone, the participant reported using the keycube eye-free for the whole case study, with the exception of very rare occasions, which were deemed accidental. The neutral keycube being deliberately devoid of character inscriptions, the visual benefit of the device was limited. Nonetheless, the first sessions were supported by the viewer highlighting key presses, reducing the need to look at the device. The user has adopted a trial-and-error (and sometimes correction) strategy maintained until the last session, including for those with the colored keycube having the character inscriptions.

Since the participant is close to expertise level, its fingers usage corroborates the results from user study 2 regarding the finger-to-key preferences (section 5.3). Seven fingers were used, with the highest load at 25.4% for the right index finger. Further details and analysis are given to the Appendix C.

10. General Discussion and Implications

Across the five studies presented in this paper, we have built up a substantial body of knowledge toward the design, usage, and performance of a tangible cubic shaped text entry interface. These studies included not only the hardware implementation of fully functional prototypes but also the development of different software tools for

the keycube and several baseline interfaces in order to validate methodologies and facilitate comparisons with the literature.

In the user study 1, a keycube (neutral, without character inscriptions) was presented to the participants to elicit the positions of the device and hands adopted while simulating text entry. The results revealed that users would prefer to adopt a position called *diagonal*: arms slightly bent with the palm of both hands touching two opposing edges of the cubic device, while having the flat surface facing down. Only from then, we could conduct the user study 2 to define the preferred and reachable keys from each finger when the cubic device is held in the preferred position *diagonal*. The results are generalizable and can be applied to other cubic-shaped devices of similar size. In addition to the literature, the results were used to design a first cubic layout based on the QWERTY, to benefit from familiarity, potentially increasing knowledge transfer. Reachability and preference scores were used to define the adaptations needed while passing from a flat surface to the cubic shape, such as favoring the use of the central (and top) face and eventually increasing the use of dexterous thumbs in comparison to the traditional keyboards.

Finally, in the user study 3, twenty-four participants typed for five 20-minute sessions with a keycube while being seated in front of a laptop. Knowledge transfer was tested and confirmed by comparing three different cubic layouts: the QWERTY-based, and two others respectively based on AZERTY (still a candidate for knowledge transfer) and DVORAK (considered unknown by the users). The performances measured included input speed and error rate as well as usability and workload induced by the device. The results show that both QWERTY and AZERTY based cubic layouts benefited from knowledge transfer, with overall performances significantly higher than the unknown cubic layout. The group of participants with the QWERTY layout reached on average 19 words per minute (wpm) within 100 minutes of training on a single half-day, which represented 30% of their input speed with a traditional keyboard. Moreover, the learning curve through the five sessions shows that the progression did not reach a plateau.

Subsequently, in the user study 4, the cubic text entry interface was tested with users wearing an augmented reality headset and standing in order to exploit the inherent mobility advantage of both devices. The study also included a comparison of performance with two baseline interfaces: the head-then-hand gesture to enter text on a mid-air virtual keyboard (HHGK), and the virtual keyboard on a smartphone (VKS) connected to the headset. Nine participants performed three 20-minute sessions, one for each interface. The main finding came by comparing the performance of the keycube session with the first session from the previous user study, the similar results showed that the keycube could be used regardless of the user posture, standing or sitting. The results also showed that the keycube performance (with the same criteria as the previous user study) was overall significantly higher than the HHGK (up to the double in input speed). Unsurprisingly, participants, who were novice with the keycube, failed to have a better input speed than VKS due to their expertise in the latter (from extensive use of the smartphone on a daily basis for many years). However, in accordance with the already exposed issues to the use of a virtual keyboard on a smartphone (VKS) connected to the headset (section 2.1), this interface received a usability score only similar to the keycube, due to its inconvenient need to look at the smartphone screen to enter text and in this case, its lack of performance progression with participants already entering text at their best. On the opposite, the keycube showed a steep progression straight from the first session and is expecting an increase of usability with more usage and expertise. Moreover, in a second part, the

study included an analysis of fingers usage while using the keycube and the results showed that after only one 20-minute session, keycube users used at least 4 fingers, up to 8, with the highest load at 36.9%. An important characteristic that shows a benefit of the keycube over the use of two-thumb text entry (e.g., a smartphone connected to a headset) prone to finger fatigue (based on repetitive muscular activities) for extensive (and professional) text entry, by overloading only one or two fingers.

At last, a longitudinal case study was conducted over a month period. The user typed two 20-sentence sessions per day, up to 40 sessions, and concluded with two additional sessions of 20 minutes in similar conditions than user studies 3 and 4. The results show not only possible features, such as eye-free capability, but also extend the current bound of user performance with a cubic shaped interface by reaching 55.1 wpm within the short period of only 215 minutes of accumulated practice. Representing 63.5% of the user’s input speed with a traditional keyboard (86.8 wpm), and an already higher performance (+8%) than the input speed with a smartphone (at 51.0 wpm). For such a touch typing user, the trade-off between the input speed or the mobility and freedom of postures is in favor of transforming the flat surface of the keyboard into a cubic shape.

The literature has shown a difference of performance between native and non-native English speakers (Palin et al., 2019). For each study reported in this paper, participants were non-native English speakers. Therefore, (1) we made sure to include relative input speed (usually to traditional keyboard) and (2) we can suggest that the absolute performance could be higher, provided the users type in their native language. Yet, despite this *disadvantage*, we believe that the overall results through the whole studies show that a keycube with known-based cubic layouts is already a practical text entry device. More importantly, over the last three studies, we have shown that users kept increasing their performance while using the keycube, unlike existing techniques, such as the virtual keyboard on smartphones, in which users already reached their best and showed no sign of improvement. For instance, in the user study 3, on a single half-day, within one and half hours of practice, users almost doubled their input speed. This continuous learning has been exacerbated in the case study, at 329% input speed progression within 3 hours of practice. The development of proficiency offered by the keycube is an important aspect for any worker already having both a deep expertise and reliance on the traditional keyboard but who is looking for more freedom of use in an immersive virtual workspace.

10.1. Implications on Workspaces

First, we recall that the concept of keycube aims to be used in a *safe* indoor environment, usually in the form of offices, either at work, at home or even while commuting (e.g., inside a train). Although possible, the keycube is not intended to be used outdoors, on the go, or for instance in an industrial context, in such cases, we believe, other solutions such as mid-air hand gesture on a virtual keyboard (Dudley et al., 2018) should be more appropriate (see the related works, part 2.1).

That being said, we embrace *The Office of the Future* depicted by Grubert, Ofek, et al. (2018) as virtual, portable and global. The authors concluded with “*We believe the freedom of the VR world along with very accurate sensing of users’ movement, their attention, and behaviors will prove to be a fertile ground for more such transformative applications that eventually will reimagine the office work as we know it*”. In that regard, the concept of keycube contributes to the direction of high freedom offered by

augmented, mixed or virtual experiences with headsets. In particular, by increasing the diversity of adopted postures with seamless transitions between them, allowing more creative and expressive work activities requiring heavy text entry. Yet, even for the post-desktop era, we do not envision offices without desktops or without seated workers, but instead, as depicted in the Figure 26, a place where workers adopt and exploit the entirety of their physical environment blended with virtual contents.



Figure 26. Examples of a keycube usage (including a neck lanyard to easily free the hands) with an augmented reality headset in an office context.

Different use cases would benefit from such approach, for instance (not limited to):

- The increasing number of collaborative tools exploiting augmented and mixed reality experiences emphasis on users standing or moving around their home and interacting with virtual avatars as in a real office, yet many of the activities in offices remain oriented toward heavy text entry. A keycube would not only decrease the burden of finding the right place to work at home, suppressing the need for a stable flat surface, but also improve the collaboration by allowing more interaction, movement, and gestures without being away from the adequate text entry interface.
- Tools requiring heavy text entry, such as an integrated development environment (IDE) could be envisioned directly while experiencing augmented and mixed reality with a headset. By doing so, fluidifying the process between developing or coding (often seated in front of a desktop computer) and testing mixed reality application (usually by moving around with a mobile device), updates and further development could be coded directly *in situ* with a keycube.
- Since, virtual content through augmented and mixed reality experiences kept increasing their presence on the society, even research papers, posters, presentations, talks and more broadly, storytelling, might widely be consumed through such mediums, improving either their understanding or entertaining. Thus, for the same reason of improving the transition and dynamics between different activities and postures, presentation programs and word processors could also benefit at being used directly from the headsets, with active users seamlessly editing their ongoing visual presentations taking advantage of the space and their accompanying text.

In addition to give more opportunities for heavy text entry activities with the headsets, we believe the concept of keycube will bring two subsequent implications:

- Sedentary office workflow in which workers remain seated for long periods has largely been reported as a major health concern (Buckley et al., 2015; Clemes, O'connell, & Edwardson, 2014) and the adoption of diverse body postures, alternating between them, have already shown benefits. The keycube could help to adopt healthier working environments by increasing the incentive (lowering the barrier) to move as well as reducing the financial and physical challenges needed

by otherwise a dedicated furniture.

- In the advent of virtual workspace being used while commuting (e.g., in a train or plane), shoulder surfing attack (SSA) from bystanders to infer keystrokes on (virtual or physical) traditional keyboards has been regarded as a serious security threat (Maiti, Jadliwala, & Weber, 2017). Thus, contrary to the relatively exposed flat surface of the traditional keyboard, a cubic shape might mitigate this privacy issue since the whole layout became impossible to observe from a single point of view, and the device could easily be hidden or moved (e.g., between the thighs when seated) while typing more confidential information.

Finally, for each conducted user study, participants notified that a keycube could be used and seemed well appropriate beyond the augmented and mixed reality experiences at work. Exemplifying more mundane contexts such as lying down in a sofa while controlling the TV or as a specific musical instrument or game controller for other media.

10.2. Limitations and Future Work

In this part, several limitations, considered as opportunities for future work, are acknowledged. We believe to have collected the major criteria needed for presenting such a novel concept. However, the traditional keyboard transformation from a plane to a cubic shape also opened a wide range of studies, in particular, since text entry is a complex task and active research domain with several specific methodologies and measures.

10.2.1. Corpus and task.

Except for the case study, the transcription was limited to alphabetical letters, space, backspace and enter. Further investigation with a corpus including numbers, special characters, upper cases and thus modifier keys remain to be done for a better generalization. For the same reason, since our user studies were limited to transcriptions, we are planning to conduct studies with composition tasks, which has been shown to be a valid complementary text entry evaluation method (Vertanen & Kristensson, 2014). We believe a keycube can stand particularly well for editing tasks, since all the known shortcuts (such as cut, copy and paste) remain transferred from the traditional keyboard, as well as moving the caret with the arrow keys (in addition to direct interactions with one hand).

10.2.2. Expertise and eye-free capability.

The case study could be reproduced with a (large) set of participants. In particular, to assess expertise through longitudinal experiments, however other methods could also be conducted such as asking participants to repeatedly type the same sentence, giving an initial but less formal expertise measure. Moreover, instead of relying on self-reports, formal methodologies should be used to explore the eye-free capability of the keycube, for instance by covering hands and the device, or by using more advanced materials to analyze eye gaze and focused areas. We believe the segmentation of keys around each face of the cube offers more tactile cues (corners and edges) and imply less hand movement than a traditional keyboard, thus, easing the eye-free capability.

10.2.3. *Virtual Reality.*

We did not investigate the keycube with headsets dedicated to virtual reality experience, yet, most of the strategies already employed to type with the traditional keyboard in virtual reality (e.g., using video-through features) could be replicated for the use of a cubic-shaped text entry interface. Moreover, in this context, eye-free capability and the use of three-dimensional virtual helpers following the orientation and movement of the real device could also be assessed.

10.2.4. *Ergonomics.*

Our ergonomic measures, particularly physical, are limited to the dimensions of the NASA-TLX (*physical demand* and *effort*), lacking accuracy (no segmentation by body members) and verbal cues (for example, adjectives qualifying the fatigue felt). We believe a keycube could have the same ergonomic benefit as a vertical keyboard, e.g., lower muscular activity, higher comfort (Muss & Hedge, 1999; Van Galen et al., 2007), an important factor for heavy text entry. Further comparative user studies remain to be done to highlight physical exertion using standardized qualitative measures such as the Borg RPE or CR10 scales (Borg, 1998) and objective measures based on physiological data (e.g., muscle activity detected with electromyography (González-Izal, Malanda, Gorostiaga, & Izquierdo, 2012)).

10.2.5. *Theoretical models.*

In addition to collecting new empirical data, theoretical models should be defined, for instance based on Hick’s Law for choice reaction and Fitts’s Law for motor performance, both already being used to predict typing speed and accuracy of traditional keyboard (Logan, Ulrich, & Lindsey, 2016). Such models could be used in addition to the data collected in the user study 2 (finger-to-key preference and reachability) and user study 4 (finger usage) to further optimize and customize the cubic layouts.

10.2.6. *Without a priori and tutorials.*

To assess the benefit of knowledge transfer, we limited the recruitment to participants with a minimal experience of traditional keyboards. Hence, we need further investigation regarding the learning curve of users without a priori, such as children. We believe the playful feature of the cubic shape could alleviate potential difficulties, inviting users to perceive it as a game, a challenge. The development of tutorials could accentuate this effect by including gamification patterns, in addition to improving the learning curve.

10.2.7. *Hardware micro-updates.*

The presented prototypes are already usable per se; however, there is room for several updates in order to reach professional commercial grade hardware and thus improve users’ performances with a keycube. For instance, the prototypes did not include the supplementary tactile cue usually presented as a bump on the letter F and J, which could serve as strong references for finger relocation. The top of all keys was flat, but concave-shaped top keys have been shown to be preferred (Ilg, 1987), preventing the fingertips from slipping and helping them to return in the center of the keys. Moreover, instead of sharing the same color per face, only specific keys could be colored with the

aim of being more visible and providing a visual landmark. The prototypes include dome switches as keys but for offering higher quality of tactile feedback, mechanical switches should be integrated, offering longer vertical travel distance as well as a range of lower and more accurate force activation.

10.2.8. Unexplored characteristics.

Lastly, some characteristics of the keycube already implemented remain unexplored in the user studies, such as vibrotactile feedback and inertial input. For instance, slight vibrations to specific key presses could improve eye-free manipulation and taking advantage of the keycube small form factor, users could shake or turn the device to switch mode, simulating toggle keys (e.g., caps lock).

11. Conclusion

The headsets dedicated to augmented and mixed reality have the potential to improve the office working environment through virtual and immersive workspaces, for instance, reducing the need for multiple displays, bringing coherence and familiarity everywhere, and finally increasing mobility and diversity of postures adopted. Yet, for such professional contexts including heavy text entry activities, relying on the traditional keyboard as the main input interface usually means bounding the workers to a limited seated posture. This paper presented and explored through a series of user studies the structural transformation from the flat surface of a keyboard to the three-dimensional cubic shape, leading to a device called keycube. By placing 80 keys around 5 faces of a cube, the handheld device maintains most of the traditional keyboard characteristics, including its usage, while bringing much higher mobility, thus being more appropriate to headsets. The key contributions of this article are in three parts. First, we present the concept and design process of the keycube with details of the hardware implementation informed by the literature. Second, we describe two user studies to elicit preferred held positions, as well as fingers-to-keys reachability and preferences in order to design a cubic layout based on QWERTY. Third, we provide empirical results through three subsequent studies, giving not only the starting learning curve of this unconventional device, but also comparative performances through a wide range of criteria for different cubic layouts, baseline interfaces and postures. We show that a keycube is a practical text entry interface in the context of virtual workplaces, bringing spatial freedom while already offering acceptable and highly progressive performance. This article establishes a useful point of reference for future studies seeking to explore a structural transformation of the traditional keyboard, if not for the cube, for another similar shape. Yet, since typing is an important generic text entry interface, a keycube could aggregate many of the strengths already exhibited by several cubic devices dedicated to different activities, and in the future, inspire other novel interfaces for virtual and immersive workspaces.

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Ethics and Disclosure statement

The authors report there are no competing interests to declare. All experiments have been approved by the University TELUQ’s Ethics Review Committee, number 2018265 and all participants voluntarily signed dedicated consent forms.

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Appendix A. Prototype exploded-view

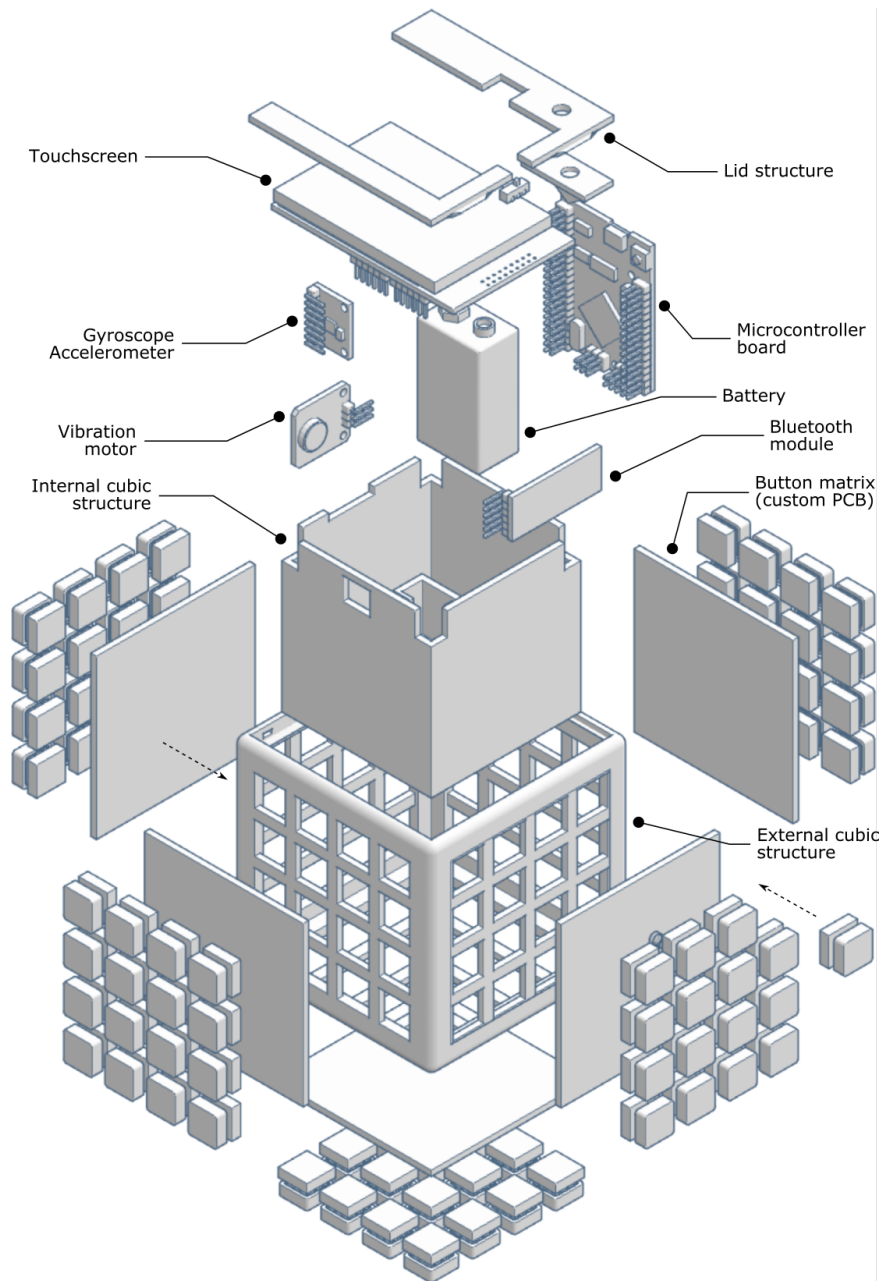


Figure A1. Detailed exploded-view from an embodiment of a keycube concept.

Appendix B. Reachability areas of the cubic shape

The results are generalizable and can be applied to other cubic-shaped devices of similar size handled in vertical position.

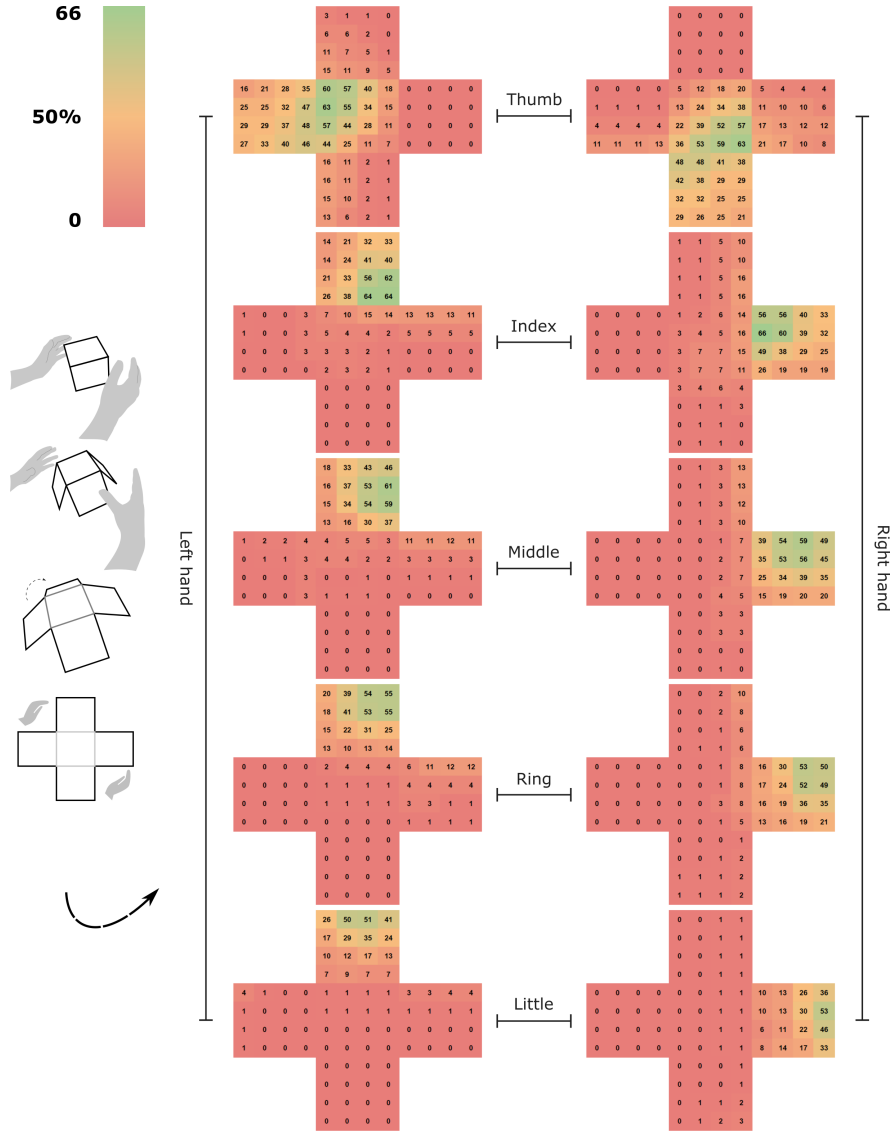


Figure B1. Reachability heat maps with scores for each finger (explanation reported in section 5.3.2).

Appendix C. Fingers Usage Further Analysis

The first section of this appendix presents a comparative analysis of the fingers usage between the cubic-shaped (keycube) and plane-shaped (traditional keyboard) devices which appened during the user study 4. The second section presents detail of the fingers usage from the longitudinal case study.

C.1. User Study 4: Augmented Reality Context and Fingers Usage

The recorded video file regarding the traditional keyboard of one participant was corrupted and one sentence from another participant was validated too early without being re-transcribed, thus, a total of 203 sentences and 6353 key presses (including correction and validation) were analyzed.

We noted two major differences between the traditional keyboard and the keycube:

- (1) The slight imbalance in hand use is reversed, with 40.7%/59.3% (left/right) for the traditional keyboard, but 58.4%/41.6% for the keycube. With a traditional keyboard, the users could adopt a strategy of moving the hands to use their right hand more frequently (considered more skillful, since all participants were right-handed) by overflowing to the left part. However, this strategy is difficult to transpose with the keycube, because the cubic shape implies a more marked separation between the left and right parts, therefore, the use of a hand for a specific key seems stricter and the rate of use ultimately reflects the theoretical left-to-right ratio calculated with respect to the frequencies of the language of the text transcribed, in our case in English, with a load of 57% for the left hand with a layout based on the QWERTY Noyes (1983b). Yet, a slight change of the space key position with the keycube (currently in the left side of the red face) could equilibrate the hand balance if desired.
- (2) A shift in the use of the fingers, from middle and index fingers for the traditional keyboard to index and thumbs for the keycube.

Despite these differences, the participants (none of them were touch typists) used mainly 4 fingers with both the traditional keyboard and the keycube, symmetrically distributed between the left and right hand. The most used fingers for the traditional keyboard and for the keycube are respectively the right middle finger (with an average usage rate of 29.3%) and the left index finger (36.3%).

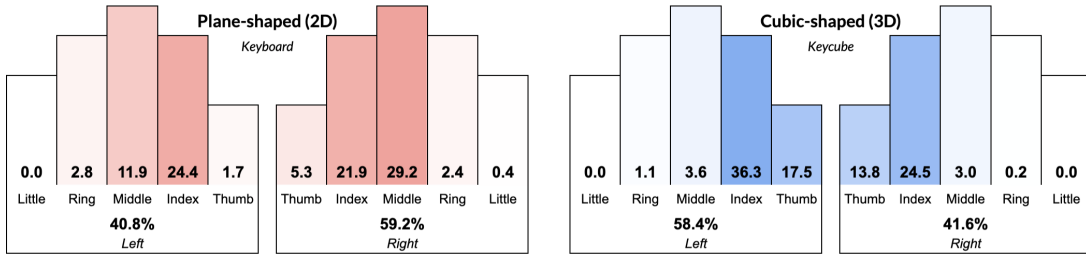


Figure C1. Fingers' usage rates for the plane (left) and cubic (right) shaped devices.

In addition, both devices offered a great deal of freedom in the choice of fingers usage to press keys, for instance some participants used mostly the middle fingers while others the index. In that regard, Figure C2 presents the usage rates of the most used finger (regardless of the finger concerned) to the least used finger among all participants. None of the participants had used more than 6 fingers with the traditional keyboard and 8 with the keycube. The result is coherent with previous studies reporting the numbers of used fingers for non-touch typists on traditional keyboard self-report at 6.95 Dhakal et al. (2018) and captured at 6.2 Feit et al. (2016). For users accustomed to typing every day with a traditional keyboard, the keycube (just after 20 minutes of practice) showed a more balanced sharing of finger usage load.

The maximum usage rate of fingers with the keycube stayed unchanged, from 36.3%

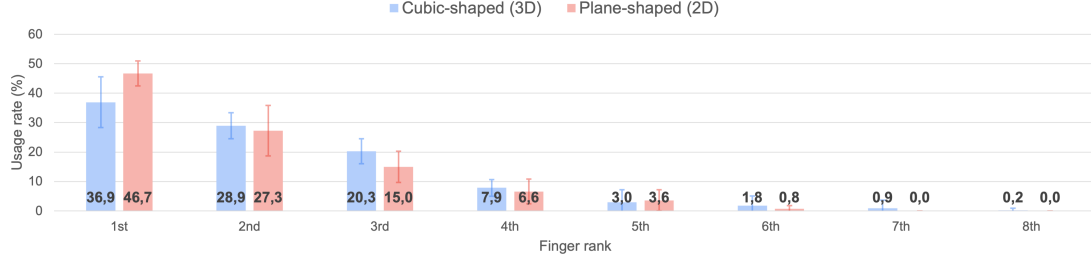


Figure C2. Usage rates from the first to the last finger used for the plane and cubic shaped devices.

(index of the left hand, Figure C1) to 36.9% (Figure C2), meaning that the fingers usage from novices was homogeneous, with one finger entering around one third of the character. However, with the traditional keyboard, the maximum usage rate of a finger increased, going from 29.3% (for the middle finger of the right hand, Figure C1) to 46.7% (Figure C2), meaning that a single finger has entered almost half of the character. This difference in fingers usage rates for the traditional keyboard is explained by the fact that the device and the experience associated with it, resulted in disparate use among participants of the finger pressing a particular key. An explanation confirmed by Figure C3 showing the usage rates of the most used finger per key across all participants for the traditional keyboard and the keycube. A rate of 100% means that the same finger was used for all the presses of the corresponding key (homogeneous usage), on the contrary a low rate (heterogeneous usage), for instance, strictly less than 50% means that there are at least three fingers used for the corresponding key. The average usage rate per key is lower with the traditional keyboard (mean=55.3, SD=11.7) than with the keycube (mean=82.4, SD=12.6). This result is particularly interesting while put into perspective with a study showing the strategies to adopt in order to improve text entry performance with a traditional keyboard Feit et al. (2016): (1) a key must be pressed by the same finger, (2) anticipating key to press and (3) limit the hand movement.

Per key, the ring fingers and little fingers were never the most used while the middle fingers are the most used only on traditional keyboard keys (mostly for the right part). We observe changes in hands on a few key presses, from the right hand with the traditional keyboard to the left hand with the keycube for the Space (Sp), but also for the keys dedicated to the letters G and V, although the latter appear to be used more with the left index finger for the traditional keyboard, the combined score of the other fingers used is slightly higher for the right hand than for the left. In addition to the purposely changes (by our design criteria) of the most used finger for the four central letters (T, Y, G, H) by switching normally from the indexes to the thumbs, the right hand is the most affected by changes of the most used fingers, mainly switching from the middle finger to the index (except the letters B and N) between the traditional keyboard and the keycube. The part dedicated to the left hand is more used with the index similarly for both devices, except for the letter Q, which is more used with the middle finger on the traditional keyboard.

C.2. Case Study: Longitudinal

The participant had none or only limited slight hand movements while using the device and shared the load of key presses over seven fingers (all but both little and right ring fingers), with the highest (index right) reaching only 25.4% (Figure C4, left). The

